

C6.2

Continuous Optimization

Lecture notes
(based on the lectures of Coralia Cartis, Oxford)

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Chapter 1

Optimality Conditions for Unconstrained Optimization

1.1 Problem classes and notation

We consider the unconstrained optimization problem

$$\min_{x \in \mathbb{R}^n} f(x), \tag{UP}$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is smooth (typically $f \in \mathcal{C}^1$ or $\mathcal{C}^2$).

Iteration notation. Iterates and other vector sequences are indexed by superscripts, e.g. x^k , s^k , u^k , δ^k , so that subscripts remain available for components such as x_i and λ_i . Scalar algorithmic parameters and named model quantities are indexed by subscripts, e.g. α_k , ε_k , Δ_k , ρ_k , B_k , m_k .

Definition 1.1 (Local and global minimizers). A point $x^* \in \mathbb{R}^n$ is a *local minimizer* of f if there exists $\delta > 0$ such that $f(x^*) \leq f(x)$ for all $x \in N(x^*, \delta) = \{x : \|x - x^*\| < \delta\}$. It is a *strict local minimizer* if the inequality is strict for $x \neq x^*$, and a *global minimizer* if $f(x^*) \leq f(x)$ for all $x \in \mathbb{R}^n$.

Definition 1.2 (Descent direction). Given $x \in \mathbb{R}^n$ with $\nabla f(x) \neq 0$, a vector $s \in \mathbb{R}^n$ is a *descent direction* for f at x if $\nabla f(x)^\top s < 0$.

Lemma 1.3 (Descent property). *If $f \in \mathcal{C}^1$ and s is a descent direction at x , then there exists $\bar{\alpha} > 0$ such that $f(x + \alpha s) < f(x)$ for all $\alpha \in (0, \bar{\alpha}]$.*

Proof. Since s is a descent direction, $\nabla f(x)^\top s < 0$. By continuity of ∇f , there exists $\bar{\alpha} > 0$ such that

$$\nabla f(x + ts)^\top s < 0$$

for all $t \in [0, \bar{\alpha}]$. Hence, for any $\alpha \in (0, \bar{\alpha}]$, by the mean value theorem there exists $t \in (0, \alpha)$ such that

$$f(x + \alpha s) - f(x) = \alpha \cdot \nabla f(x + ts)^\top s < 0.$$

□

1.2 First-order optimality conditions

Theorem 1.4 (First-order necessary conditions). *Let $f \in \mathcal{C}^1(\mathbb{R}^n)$. If x^* is a local minimizer of f , then $\nabla f(x^*) = 0$.*

Proof. Suppose not. Put $s = -\nabla f(x^*)$ and note that s is a descent direction. By Lemma 1.3 there exists $\bar{\alpha} > 0$ such that $f(x^* + \alpha s) < f(x^*)$ for all $\alpha \in (0, \bar{\alpha}]$, contradiction. $\square$

Definition 1.5 (Stationary point). A point x with $\nabla f(x) = 0$ is called a *stationary point* of f .

1.3 Second-order optimality conditions

Theorem 1.6 (Second-order necessary conditions). *Let $f \in \mathcal{C}^2(\mathbb{R}^n)$. If x^* is a local minimizer of f , then $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \succeq 0$ (positive semidefinite), i.e. $s^\top \nabla^2 f(x^*) s \geq 0$ for all $s \in \mathbb{R}^n$.*

Proof. By Theorem 1.4, $\nabla f(x^*) = 0$. Suppose for contradiction that $s^\top \nabla^2 f(x^*) s < 0$ for some $s \in \mathbb{R}^n$. Since $s \neq 0$ and $f \in \mathcal{C}^2$, there exists $\hat{\alpha} > 0$ such that $s^\top \nabla^2 f(x^* + \alpha s) s < 0$ for all $\alpha \in (0, \hat{\alpha}]$. By Taylor's theorem, for any $\alpha \in (0, \hat{\alpha}]$, there exists $t \in (0, \alpha)$ such that

$$f(x^* + \alpha s) - f(x^*) = \frac{\alpha^2}{2} s^\top \nabla^2 f(x^* + ts) s < 0,$$

contradiction. $\square$

Theorem 1.7 (Second-order sufficient conditions). *Let $f \in \mathcal{C}^2(\mathbb{R}^n)$. If $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \succ 0$ (positive definite), then x^* is a strict local minimizer of f .*

Proof. Since $\nabla^2 f(x^*) \succ 0$ and $\nabla^2 f$ is continuous, there exists $\delta > 0$ such that $\nabla^2 f(x) \succ 0$ for all $x \in N(x^*, \delta)$. Now let $0 \neq d \in \mathbb{R}^n$ with $\|d\| < \delta$. By Taylor's theorem with $\nabla f(x^*) = 0$, there exists $\tilde{t} \in (0, 1)$ such that

$$f(x^* + d) - f(x^*) = \nabla f(x^*)^\top d + \frac{1}{2} d^\top \nabla^2 f(x^* + \tilde{t} d) d = \frac{1}{2} d^\top \nabla^2 f(x^* + \tilde{t} d) d > 0,$$

since $\|\tilde{t} d\| < \delta$, so $\nabla^2 f(x^* + \tilde{t} d) \succ 0$. Thus $f(x) > f(x^*)$ for all $x \in N(x^*, \delta) \setminus \{x^*\}$. $\square$

Example 1.8 (Necessary $\nRightarrow$ sufficient). The conditions are different:

- $f(x) = x^3$, $x^* = 0$: satisfies the second-order necessary condition ($f''(0) = 0 \geq 0$) but is not a local minimizer.
- $f(x) = x^4$, $x^* = 0$: is a strict local minimizer but the second-order sufficient condition *fails* ($f''(0) = 0 \not\succ 0$).

The gap between necessary and sufficient is the case $\nabla^2 f(x^*) \succeq 0$ but not $\succ 0$, in which case optimality cannot be decided from ∇f and $\nabla^2 f$ alone.

1.4 Convex problems

For convex problems, the picture simplifies dramatically: *every local minimizer is global*, and the first-order necessary condition $\nabla f(x^*) = 0$ becomes *sufficient*. This is one of the main reasons convex optimization is algorithmically tractable.

Definition 1.9 (Convex function). A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is *convex* if its domain is convex and, for all x, y in its domain and all $\lambda \in [0, 1]$, $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$.

Theorem 1.10 (Optimality for convex problems). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex and $\mathcal{C}^1$. Then x^* is a global minimizer of f if and only if $\nabla f(x^*) = 0$.*

Proof. If x^* is a global minimizer, then by Theorem 1.4, $\nabla f(x^*) = 0$. It remains to show the converse: $\nabla f(x^*) = 0$ implies x^* is a global minimizer.

Since f is convex and $\mathcal{C}^1$, the first-order characterisation of convexity gives

$$f(y) \geq f(x^*) + \nabla f(x^*)^\top (y - x^*) = f(x^*) \quad \text{for all } y \in \mathbb{R}^n,$$

using $\nabla f(x^*) = 0$. Hence x^* is a global minimizer. □

Corollary 1.11 (Local = global for convex problems). *If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and x^* is a local minimizer of f , then x^* is a global minimizer.*

Proof. Suppose x^* is a local minimizer but not a global minimizer. Then there exists y with $f(y) < f(x^*)$. By convexity, for any $\lambda \in (0, 1)$,

$$f(x^* + \lambda(y - x^*)) \leq (1 - \lambda)f(x^*) + \lambda f(y) < f(x^*).$$

Taking λ small enough that $\|\lambda(y - x^*)\| < \delta$ contradicts local minimality. □

Chapter 2

Linesearch Methods

A linesearch method generates iterates $x^{k+1} = x^k + \alpha_k s^k$ where s^k is a descent direction and $\alpha_k > 0$ is a stepsize. The philosophy is “liberal in the choice of search direction, keeping bad behaviour in control by choice of α_k ”. Choosing α_k properly is essential: too long and the method may overshoot, too short and progress is negligible.

2.1 The generic linesearch method

Algorithm 2.1 (Generic Linesearch Method (GLM)). Choose $\varepsilon > 0$ and $x^0 \in \mathbb{R}^n$. For $k \geq 0$:

1. If $\|\nabla f(x^k)\| \leq \varepsilon$, stop.
2. Compute a descent direction s^k with $\nabla f(x^k)^\top s^k < 0$.
3. Compute a stepsize $\alpha_k > 0$ along s^k such that $f(x^k + \alpha_k s^k) < f(x^k)$.
4. Set $x^{k+1} := x^k + \alpha_k s^k$.

2.2 Stepsize rules: the Armijo condition

Choices of α_k . The conceptually simplest choice is the *exact linesearch* $\alpha_k := \arg \min_{\alpha > 0} f(x^k + \alpha s^k)$, but this is computationally expensive for nonlinear f . Inexact linesearches replace it with a cheap rule that guarantees the stepsize is “not too long” and “not too short”.

The *Armijo condition* demands sufficient decrease: α is acceptable provided the actual decrease is at least a fraction β of what the linear model predicts. In practice β is taken small, e.g. $\beta = 0.1$ or even $\beta = 0.001$, to keep the rule mild.

For $\beta \in (0, 1)$ the *Armijo (sufficient decrease) condition* at iteration k is

$$f(x^k + \alpha s^k) \leq f(x^k) + \beta \alpha \nabla f(x^k)^\top s^k. \quad (\text{ac})$$

2.3 The Wolfe conditions

The Armijo condition prevents α_k from being too long, but on its own admits arbitrarily small α_k . The *Wolfe conditions* add a second (“curvature”) inequality that rules out steps that are too short. Wolfe linesearches are needed for some methods (e.g. BFGS, see Chapter 5) to ensure the curvature condition $(\delta^k)^\top \gamma^k > 0$.

Definition 2.2 (Wolfe conditions). Given $0 < \beta < \eta < 1$, the stepsize $\alpha > 0$ satisfies the *Wolfe conditions* if

$$f(x^k + \alpha s^k) \leq f(x^k) + \beta \alpha \nabla f(x^k)^\top s^k, \quad (\text{W1})$$

$$\nabla f(x^k + \alpha s^k)^\top s^k \geq \eta \nabla f(x^k)^\top s^k. \quad (\text{W2})$$

Lemma 2.3 (Existence of an Armijo stepsize). *Let $f \in \mathcal{C}^1$, let s^k be a descent direction at x^k , and let $\beta \in (0, 1)$. Then there exists $\bar{\alpha} > 0$ such that (W1) holds for all $\alpha \in [0, \bar{\alpha}]$.*

Proof. Define $\varphi(\alpha) := f(x^k + \alpha s^k)$. Then $\varphi(0) = f(x^k)$ and $\varphi'(0) = \nabla f(x^k)^\top s^k < 0$. Since $f \in \mathcal{C}^1$, φ' is continuous. But then the set $\{\alpha : \varphi'(\alpha) < \beta \varphi'(0)\}$ is open and contains 0, so there exists $\bar{\alpha} > 0$ such that

$$\varphi'(\alpha) < \beta \varphi'(0) \quad \text{for all } \alpha \in [0, \bar{\alpha}].$$

By the mean value theorem, for each $\alpha \in (0, \bar{\alpha}]$, there exists $\tilde{\alpha} \in (0, \alpha)$ with

$$\varphi(\alpha) = \varphi(0) + \alpha \varphi'(\tilde{\alpha}) < \varphi(0) + \beta \alpha \varphi'(0),$$

which is exactly the Armijo condition (W1). $\square$

Lemma 2.4 (Existence of a Wolfe stepsize). *Let $f \in \mathcal{C}^1$ be bounded below, let s^k be a descent direction at x^k , and let $0 < \beta < \eta < 1$. Then there exists $\alpha^* > 0$ satisfying both (W1) and (W2).*

Proof. Define $\varphi(\alpha) := f(x^k + \alpha s^k)$, so $\varphi'(\alpha) = \nabla f(x^k + \alpha s^k)^\top s^k$. The Armijo line is $\ell(\alpha) := \varphi(0) + \beta \alpha \varphi'(0)$. Since $\ell(\alpha) \rightarrow -\infty$ as $\alpha \rightarrow \infty$ and $\varphi(\alpha) = f(x^k + \alpha s^k)$ is bounded below, the Armijo condition $\varphi(\alpha) \leq \ell(\alpha)$ cannot hold for all $\alpha > 0$. By Lemma 2.3 it holds for small α , so there exists a smallest $\bar{\alpha} > 0$ at which it first fails, i.e. $\varphi(\bar{\alpha}) = \ell(\bar{\alpha})$ and $\varphi(\alpha) \leq \ell(\alpha)$ for $\alpha \in [0, \bar{\alpha}]$.

By the mean value theorem applied to φ on $[0, \bar{\alpha}]$, there exists $\alpha^* \in (0, \bar{\alpha})$ with

$$\varphi'(\alpha^*) = \frac{\varphi(\bar{\alpha}) - \varphi(0)}{\bar{\alpha}} = \frac{\ell(\bar{\alpha}) - \varphi(0)}{\bar{\alpha}} = \beta \varphi'(0) > \eta \varphi'(0)$$

Thus (W2) holds at α^* . And $\alpha^* < \bar{\alpha}$, so $\varphi(\alpha^*) \leq \ell(\alpha^*)$, i.e. (W1) holds as well. $\square$

Algorithm 2.5 (Backtracking-Armijo (bArmijo) linesearch). Choose $\alpha^{(0)} > 0$, $\tau \in (0, 1)$, $\beta \in (0, 1)$. Set $i := 0$. While (ac) fails with $\alpha = \alpha^{(i)}$, set $\alpha^{(i+1)} := \tau \alpha^{(i)}$ and $i := i + 1$. Set $\alpha_k := \alpha^{(i)}$.

Lemma 2.6 (Descent lemma). *Let $f \in \mathcal{C}^1$ with ∇f Lipschitz continuous with constant L . Then for any $x, s \in \mathbb{R}^n$ and $\alpha > 0$,*

$$f(x + \alpha s) \leq f(x) + \alpha \nabla f(x)^\top s + \alpha^2 L \|s\|^2.$$

Proof. By Taylor's theorem, there exists $\tilde{\alpha} \in (0, \alpha)$ such that $f(x + \alpha s) = f(x) + \alpha \nabla f(x + \tilde{\alpha} s)^\top s$. Adding and subtracting $\nabla f(x)$, then applying Cauchy-Schwarz and the Lipschitz condition:

$$\begin{aligned} f(x + \alpha s) &= f(x) + \alpha \nabla f(x)^\top s + \alpha [\nabla f(x + \tilde{\alpha} s) - \nabla f(x)]^\top s \\ &\leq f(x) + \alpha \nabla f(x)^\top s + \alpha \|\nabla f(x + \tilde{\alpha} s) - \nabla f(x)\| \|s\| \\ &\leq f(x) + \alpha \nabla f(x)^\top s + \alpha L \tilde{\alpha} \|s\|^2 \\ &\leq f(x) + \alpha \nabla f(x)^\top s + \alpha^2 L \|s\|^2. \end{aligned} \quad \square$$

Lemma 2.7 (Range of admissible stepsizes). *Let $f \in \mathcal{C}^1$ with ∇f Lipschitz continuous with constant L . At iteration k of GLM, the Armijo condition (ac) is satisfied for all $\alpha \in [0, \alpha_{k,\max}]$, where*

$$\alpha_{k,\max} = \frac{(\beta - 1) \nabla f(x^k)^\top s^k}{L \|s^k\|^2}.$$

Proof. First note that $\alpha_{k,\max} > 0$ as $\beta < 1$ and s^k is a descent direction. Let s^k be a descent direction and let $\alpha \in [0, \alpha_{k,\max}]$. By Lemma 2.6,

$$\begin{aligned} f(x^k + \alpha s^k) &\leq f(x^k) + \alpha \nabla f(x^k)^\top s^k + \alpha^2 L \|s^k\|^2 \\ &\leq f(x^k) + \alpha \nabla f(x^k)^\top s^k + \alpha(\beta - 1) \nabla f(x^k)^\top s^k \\ &\leq f(x^k) + \beta \alpha \nabla f(x^k)^\top s^k, \end{aligned}$$

which satisfies the Armijo condition. $\square$

Lemma 2.8 (Lower bound on bArmijo stepsize). *Under the conditions of Lemma 2.7, the bArmijo stepsize α_k satisfies $\alpha_k \geq \min\{\alpha^{(0)}, \tau \alpha_{k,\max}\}$ for all $k \geq 0$.*

Proof. Note that $\alpha_k = \alpha^{(0)}$ if bArmijo terminates at the start. Suppose this is not the case. By Lemma 2.7, bArmijo terminates whenever $\alpha \leq \alpha_{k,\max}$. Let $i - 1$ be the last iteration before termination, so $\alpha^{(i-1)} > \alpha_{k,\max}$ and $\tau \alpha^{(i-1)} = \alpha^{(i)} \leq \alpha_{k,\max}$. But then

$$\alpha_k = \alpha^{(i)} = \tau \alpha^{(i-1)} > \tau \alpha_{k,\max}. \quad \square$$

2.4 Global convergence of GLM

Theorem 2.9 (Global convergence of GLM with bArmijo). *Let $f \in \mathcal{C}^1(\mathbb{R}^n)$ be bounded below by f_{low} , and ∇f be Lipschitz continuous. Apply GLM with bArmijo linesearch ($\varepsilon := 0$). Then either there exists $\ell \geq 0$ with $\nabla f(x^\ell) = 0$, or*

$$\lim_{k \rightarrow \infty} \min \left\{ \frac{|\nabla f(x^k)^\top s^k|}{\|s^k\|}, |\nabla f(x^k)^\top s^k| \right\} = 0.$$

Proof. Suppose $\nabla f(x^k) \neq 0$ for all $k \geq 0$. Then Armijo condition with $x^{k+1} = x^k + \alpha_k s^k$ gives

$$f(x^{k+1}) \leq f(x^k) + \beta \alpha_k \nabla f(x^k)^\top s^k \implies f(x^k) - f(x^{k+1}) \geq \beta \alpha_k |\nabla f(x^k)^\top s^k|.$$

for all $k \geq 0$. By the telescoping sum,

$$f(x^0) - f(x^{k+1}) \geq \sum_{i=0}^k \beta \alpha_i |\nabla f(x^i)^\top s^i|.$$

Since f is bounded below by f_{low} ,

$$\infty > f(x^0) - f_{\text{low}} \geq \sum_{i=0}^{\infty} \beta \alpha_i |\nabla f(x^i)^\top s^i|.$$

Thus, $\alpha_k |\nabla f(x^k)^\top s^k| \rightarrow 0$ as $k \rightarrow \infty$. Recall from Lemma 2.8 that $\alpha_k \geq \min\{\alpha^{(0)}, \tau \alpha_{k,\max}\}$ for all $k \geq 0$. Define

$$\mathcal{K}_1 := \{k : \alpha^{(0)} \geq \tau \alpha_{k,\max}\}, \quad \mathcal{K}_2 := \{k : \alpha^{(0)} < \tau \alpha_{k,\max}\}.$$

For $k \in \mathcal{K}_1$,

$$\alpha_k |\nabla f(x^k)^\top s^k| \geq \frac{(1-\beta)\tau}{L} \cdot \left(\frac{|\nabla f(x^k)^\top s^k|}{\|s^k\|} \right)^2 \geq 0.$$

But then $\alpha_k |\nabla f(x^k)^\top s^k| \rightarrow 0$ as $k \rightarrow \infty$ in $\mathcal{K}_1$, so

$$\lim_{\substack{k \rightarrow \infty \\ k \in \mathcal{K}_1}} \frac{|\nabla f(x^k)^\top s^k|}{\|s^k\|} \rightarrow 0.$$

On the other hand, for $k \in \mathcal{K}_2$ we have $\alpha_k \geq \alpha^{(0)}$, so

$$0 \leq \lim_{\substack{k \rightarrow \infty \\ k \in \mathcal{K}_2}} |\nabla f(x^k)^\top s^k| \leq \frac{1}{\alpha^{(0)}} \cdot \lim_{\substack{k \rightarrow \infty \\ k \in \mathcal{K}_2}} \alpha_k |\nabla f(x^k)^\top s^k| \rightarrow 0.$$

Combining both cases, we have

$$\lim_{k \rightarrow \infty} \min \left\{ \frac{|\nabla f(x^k)^\top s^k|}{\|s^k\|}, |\nabla f(x^k)^\top s^k| \right\} = 0.$$

□

Remark 2.10. Writing $\cos \theta_k = \frac{-\nabla f(x^k)^\top s^k}{\|\nabla f(x^k)\| \|s^k\|}$, Theorem 2.9 gives $\|\nabla f(x^k)\| \cos \theta_k \min\{1, \|s^k\|\} \rightarrow 0$. Hence to ensure $\|\nabla f(x^k)\| \rightarrow 0$ it is not enough that s^k be descent: the angle condition $\cos \theta_k \geq \delta > 0$ for some δ is required.

Chapter 3

The Steepest Descent Method

The steepest descent (SD) method is the simplest concrete instance of GLM, obtained by choosing $s^k = -\nabla f(x^k)$. It is a *first-order method*: only gradient information is used.

3.1 The steepest descent direction

Definition 3.1 (Steepest descent direction). The *steepest descent (SD) direction* at x^k is $s_{\text{SD}}^k := -\nabla f(x^k)$.

Lemma 3.2 (Optimality of $-\nabla f$). *The steepest descent direction is the unique global solution of*

$$\min_{s \in \mathbb{R}^n} f(x^k) + s^\top \nabla f(x^k) \quad \text{subject to} \quad \|s\| = \|\nabla f(x^k)\|.$$

Proof. Note that

$$\arg \min_{s \in \mathbb{R}^n} f(x^k) + s^\top \nabla f(x^k) = \arg \min_{s \in \mathbb{R}^n} s^\top \nabla f(x^k).$$

By Cauchy-Schwarz inequality,

$$|s^\top \nabla f(x^k)| \leq \|s\| \|\nabla f(x^k)\| = \|\nabla f(x^k)\|^2,$$

with equality if and only if $s = \pm \nabla f(x^k)$. In particular,

$$s^\top \nabla f(x^k) \geq -\|\nabla f(x^k)\|^2,$$

which is only achieved when $s = -\nabla f(x^k)$. □

3.2 Global convergence of steepest descent

Theorem 3.3 (Global convergence of SD with bArmijo). *Let $f \in \mathcal{C}^1(\mathbb{R}^n)$ be bounded below with ∇f Lipschitz continuous. Apply GLM with $s^k = -\nabla f(x^k)$ and bArmijo linesearch. Then either there exists ℓ with $\nabla f(x^\ell) = 0$, or $\|\nabla f(x^k)\| \rightarrow 0$ as $k \rightarrow \infty$.*

Proof. Suppose there are no ℓ with $\nabla f(x^\ell) = 0$. Then by Theorem 2.9,

$$\lim_{k \rightarrow \infty} \min \left\{ \frac{\|\nabla f(x^k)\|^2}{\|-\nabla f(x^k)\|}, \|\nabla f(x^k)\| \right\} = \|\nabla f(x^k)\| = 0. \quad \square$$

Corollary 3.4 (Iteration complexity of SD with bArmijo). *Under the conditions of Theorem 3.3, for any $\varepsilon > 0$, there exists an iteration k with $\|\nabla f(x^k)\| \leq \varepsilon$ and*

$$k \leq \frac{f(x^0) - f_{\text{low}}}{C \varepsilon^2}, \quad C := \beta \min \left\{ \alpha^{(0)}, \frac{(1 - \beta)\tau}{L} \right\},$$

where L is the Lipschitz constant of ∇f , β is the Armijo parameter, τ the backtracking factor, and $\alpha^{(0)}$ the initial stepsize.

Proof. For steepest descent with $s^k = -\nabla f(x^k)$, Lemma 2.8 gives $\alpha_k \geq \min\{\alpha^{(0)}, \tau(1 - \beta)/L\} =: C'$. The Armijo condition then yields

$$f(x^k) - f(x^{k+1}) \geq \beta \alpha_k \|\nabla f(x^k)\|^2 \geq \beta C' \|\nabla f(x^k)\|^2 = C \|\nabla f(x^k)\|^2.$$

If $\|\nabla f(x^i)\| > \varepsilon$ for all $i = 0, \dots, k - 1$, telescoping gives

$$f(x^0) - f_{\text{low}} \geq \sum_{i=0}^{k-1} C \varepsilon^2 = k C \varepsilon^2,$$

so $k \leq (f(x^0) - f_{\text{low}})/(C \varepsilon^2)$. □

Remark 3.5. The $O(1/\varepsilon^2)$ complexity is a worst-case bound for first-order methods on smooth nonconvex problems. Newton's method can be faster asymptotically, but the $O(1/\varepsilon^2)$ bound is tight for steepest descent without further assumptions.

3.3 Local rate of convergence

Definition 3.6 (Rates of convergence). A sequence $\{x^k\} \subset \mathbb{R}^n$ converges to x^* with rate $p \geq 1$ if there exist $\rho > 0$ and k_0 with $\|x^{k+1} - x^*\| \leq \rho \|x^k - x^*\|^p$ for all $k \geq k_0$. The cases $p = 1$ ($\rho < 1$) and $p = 2$ are called *linear* and *quadratic* convergence respectively. The convergence is *superlinear* if $\|x^{k+1} - x^*\| / \|x^k - x^*\| \rightarrow 0$.

Theorem 3.7 (Local linear convergence of SD). *Let $f \in \mathcal{C}^2$ and let x^* be a local minimizer of f with $\nabla^2 f(x^*) \succ 0$ and eigenvalues $0 < \lambda_{\min}^* \leq \lambda_{\max}^*$. Apply SD with exact linesearch. If $x^k \rightarrow x^*$, then the convergence is linear with factor*

$$\rho \leq \frac{\kappa(x^*) - 1}{\kappa(x^*) + 1}, \quad \kappa(x^*) = \frac{\lambda_{\max}^*}{\lambda_{\min}^*}.$$

□

3.4 Summary of steepest descent

- **Inexpensive:** only first derivatives required.
- **Strong global convergence** under weak assumptions, with second-order optimality guarantees for the generated solution (see e.g. saddle-point avoidance properties in Lecture 5).
- **Scale-dependent:** a poor scaling of variables produces a steepest-descent direction that gives little progress. Newton's method does not have this defect (Theorem 4.4).

- **Slow asymptotic convergence:** when $\kappa(x^*)$ is large, $\rho_{SD} = (\kappa - 1)/(\kappa + 1)$ is very close to 1 and the iterates “zig-zag”. SD is therefore not used in general-purpose software, but it remains useful for certain very well-conditioned problems with structure (e.g. compressed sensing).

Chapter 4

Newton's Method

Newton's method for unconstrained minimization can be derived in two equivalent ways:

1. *Quadratic model.* The Newton step s^k minimizes the second-order Taylor approximation of f at x^k (when the Hessian is positive definite). Compare this with steepest descent, whose direction minimizes the first-order Taylor approximation under a norm constraint.
2. *Newton's root-finding.* Stationarity is the equation $\nabla f(x) = 0$, an $n \times n$ nonlinear system. Applying Newton's root-finding scheme to $r(x) := \nabla f(x)$ produces $x^{k+1} = x^k - [J(x^k)]^{-1}r(x^k) = x^k - [\nabla^2 f(x^k)]^{-1}\nabla f(x^k)$, recovering the Newton iterate.

4.1 The Newton direction

Definition 4.1 (Newton direction). For $f \in \mathcal{C}^2$ with $\nabla^2 f(x^k)$ nonsingular, the *Newton direction* at x^k is the solution s^k of

$$\nabla^2 f(x^k) s^k = -\nabla f(x^k). \quad (4.1)$$

The (pure) Newton iteration sets $x^{k+1} := x^k + s^k$.

Lemma 4.2 (Descent property of the Newton direction). If $\nabla^2 f(x^k) \succ 0$ and $\nabla f(x^k) \neq 0$, then the Newton direction is a descent direction. Moreover, s^k is the unique minimizer of the second-order Taylor model $q_k(s) := f(x^k) + s^\top \nabla f(x^k) + \frac{1}{2}s^\top \nabla^2 f(x^k)s$.

Proof. Put $H_k := \nabla^2 f(x^k)$ and $g_k := \nabla f(x^k)$. Since $H_k \succ 0$, it is nonsingular and the Newton equation gives

$$s^k = -H_k^{-1}g_k.$$

Also $H_k^{-1} \succ 0$, so, as $g_k \neq 0$,

$$g_k^\top s^k = -g_k^\top H_k^{-1}g_k < 0,$$

which makes s^k a descent direction.

It remains to prove the model-minimizing property. Completing the square, we may write

$$q_k(s) = f(x^k) - \frac{1}{2}g_k^\top H_k^{-1}g_k + \frac{1}{2}(s + H_k^{-1}g_k)^\top H_k(s + H_k^{-1}g_k).$$

The last term is nonnegative, and it vanishes if and only if $s = -H_k^{-1}g_k = s^k$. Hence s^k is the unique global minimizer of q_k . $\square$

4.2 Local convergence of pure Newton

Theorem 4.3 (Local quadratic convergence of pure Newton). *Let $f \in \mathcal{C}^2(\mathbb{R}^n)$, $\nabla f(x^*) = 0$, $\nabla^2 f(x^*)$ nonsingular, and $\nabla^2 f$ locally Lipschitz continuous at x^* . If x^{k_0} is sufficiently close to x^* for some $k_0 \geq 0$, then x^k is well-defined for all $k \geq k_0$ and $x^k \rightarrow x^*$ at a quadratic rate.*

Proof. Write $g(x) := \nabla f(x)$ and $H(x) := \nabla^2 f(x)$. Since $H(x^*)$ is nonsingular and locally Lipschitz continuous and H is continuous, there is a radius $\rho > 0$ such that for $x, y \in N(x^*, \rho)$,

- $H(x)^{-1}$ exists and $\|H(x)^{-1}\| \leq M$ for some $M > 0$,
- $\|H(y) - H(x)\| \leq L \|y - x\|$.

For $x \in N(x^*, \rho)$, Taylor's theorem gives

$$g(x^*) = g(x) + H(x)(x^* - x) + r(x),$$

where the remainder term

$$\begin{aligned} \|r(x)\| &= \|g(x+h) - g(x) - H(x)h\| \\ &\leq \int_0^1 \|H(x+th) - H(x)\| \cdot \|h\| dt \\ &\leq L \|h\|^2 \int_0^1 t dt = \frac{L}{2} \|x^* - x\|^2, \end{aligned}$$

by Lipschitz and $h := x^* - x$.

Since $g(x^*) = 0$, the pure Newton iterate $N(x) := x - H(x)^{-1}g(x)$ satisfies

$$N(x) - x^* = H(x)^{-1}(H(x)(x - x^*) - g(x)) = H(x)^{-1}r(x).$$

Hence

$$\|N(x) - x^*\| \leq \frac{ML}{2} \|x - x^*\|^2.$$

Choose $\delta \in (0, \rho)$ so small that $\frac{ML}{2}\delta \leq \frac{1}{2}$. If $x^{k_0} \in N(x^*, \delta)$, the above estimate shows inductively that every Newton iterate is well-defined and remains in $N(x^*, \delta)$, since

$$\|x^{k+1} - x^*\| \leq \frac{ML}{2} \|x^k - x^*\|^2 \leq \frac{1}{2} \|x^k - x^*\|.$$

Therefore, $x^k \rightarrow x^*$ and the first inequality gives quadratic convergence. $\square$

Theorem 4.4 (Scale invariance of Newton). *Newton's method (with or without linesearch) is scale invariant under linear transformations of the variables. Explicitly, if $A \in \mathbb{R}^{n \times n}$ is nonsingular, $y = Ax$, and $\bar{f}(y) := f(A^{-1}y)$, then the Newton direction s_y for $\bar{f}$ at y satisfies $y + \alpha s_y = A(x + \alpha s_x)$.*

Proof. Note that

$$\nabla \bar{f}(y) = (A^{-1})^\top \nabla f(x), \quad \nabla^2 \bar{f}(y) = (A^{-1})^\top \nabla^2 f(x) A^{-1}.$$

Thus,

$$s_y = -[\nabla^2 \bar{f}(y)]^{-1} \nabla \bar{f}(y) = (A[\nabla^2 f(x)]^{-1} A^\top)((A^{-1})^\top \nabla f(x)) = A s_x.$$

$\square$

Remark 4.5 (Disadvantages of Newton's method). Despite its fast local convergence, the pure Newton iteration suffers several serious drawbacks:

- s^k is not well-defined when $\nabla^2 f(x^k)$ is singular.
- s^k may not be a descent direction when $\nabla^2 f(x^k)$ is not positive definite.
- Pure Newton may be attracted to local maxima or saddle points (the local convergence Theorem 4.3 only requires $\nabla^2 f(x^*)$ nonsingular, not positive definite).
- Pure Newton may fail to converge if x^0 is far from a solution; for example, applied to $f(x) = -x^6/6 + x^4/4 + 2x^2$ from $x^0 = 1$, it cycles between ± 1 (neither stationary).

This is why a linesearch (or trust-region) safeguard is needed to make Newton's method globally convergent.

4.3 Linesearch Newton (damped Newton) method

Algorithm 4.6 (Linesearch Newton method). Choose $\varepsilon > 0$ and $x^0 \in \mathbb{R}^n$. While $\|\nabla f(x^k)\| > \varepsilon$: solve $\nabla^2 f(x^k)s^k = -\nabla f(x^k)$, then set $x^{k+1} = x^k + \alpha_k s^k$ with $\alpha_k \in (0, 1]$ chosen by, e.g., bArmijo linesearch.

Theorem 4.7 (Local convergence of linesearch Newton). *Let $f \in \mathcal{C}^2(\mathbb{R}^n)$ with $\nabla^2 f$ Lipschitz continuous and positive definite at the iterates. Apply Newton's method with bArmijo linesearch using $\beta < 1/2$ and $\alpha^{(0)} = 1$. If $x^k \rightarrow x^*$ with $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \succ 0$, then $\alpha_k = 1$ for all k sufficiently large and the convergence is asymptotically quadratic.*

Proof. Put $g_k := \nabla f(x^k)$, $H_k := \nabla^2 f(x^k)$, and let $s^k = -H_k^{-1}g_k$ be the Newton direction. Since $x^k \rightarrow x^*$ and $H(x^*) \succ 0$, there exist $m > 0$ and k_0 such that

$$H_k \succeq mI \quad \text{for all } k \geq k_0.$$

In particular, for $k \geq k_0$,

$$g_k^\top s^k = -(s^k)^\top H_k s^k \leq -m \|s^k\|^2.$$

Also $g_k \rightarrow 0$ and H_k^{-1} is bounded for large k , so $s^k \rightarrow 0$.

Let L be a Lipschitz constant for $\nabla^2 f$ on a neighbourhood of x^* containing x^k and $x^k + s^k$ for all sufficiently large k . Taylor's theorem with Lipschitz Hessian gives

$$f(x^k + s^k) \leq f(x^k) + g_k^\top s^k + \frac{1}{2}(s^k)^\top H_k s^k + \frac{L}{6} \|s^k\|^3.$$

Using $H_k s^k = -g_k$, this becomes

$$f(x^k + s^k) \leq f(x^k) + \frac{1}{2} g_k^\top s^k + \frac{L}{6} \|s^k\|^3.$$

Since $\beta < 1/2$ and $s^k \rightarrow 0$, for all sufficiently large k ,

$$\frac{L}{6} \|s^k\| \leq \left(\frac{1}{2} - \beta\right) m.$$

Therefore

$$\frac{L}{6} \|s^k\|^3 \leq \left(\frac{1}{2} - \beta\right) (s^k)^\top H_k s^k = \left(\frac{1}{2} - \beta\right) (-g_k^\top s^k),$$

and hence

$$f(x^k + s^k) \leq f(x^k) + \beta g_k^\top s^k.$$

Thus the full step $\alpha = 1$ satisfies the Armijo condition. Since bArmijo starts with $\alpha^{(0)} = 1$, it accepts $\alpha_k = 1$ for all sufficiently large k .

From that point on the method is pure Newton. The hypotheses of Theorem 4.3 hold at x^* , so the tail of the sequence satisfies

$$\|x^{k+1} - x^*\| \leq C \|x^k - x^*\|^2$$

for some constant $C > 0$ and all sufficiently large k . This is asymptotic quadratic convergence. $\square$

4.4 Global convergence with general second-order models

In GLM, let B_k be symmetric positive definite for all k , and let s^k solve

$$B_k s^k = -\nabla f(x^k). \quad (4.2)$$

Theorem 4.8 (Global convergence with bounded B_k). *Let $f \in \mathcal{C}^1(\mathbb{R}^n)$ be bounded below and ∇f Lipschitz continuous. Apply GLM with s^k from (4.2) and bArmijo linesearch ($\varepsilon := 0$). Suppose the eigenvalues of B_k are uniformly bounded above and uniformly bounded away from zero, independently of k . Then either there exists ℓ with $\nabla f(x^\ell) = 0$, or $\|\nabla f(x^k)\| \rightarrow 0$ as $k \rightarrow \infty$.*

Proof. Write $H_k = \nabla^2 f(x^k)$ and $g_k = \nabla f(x^k)$. Since $B_k \succ 0$, we may write $s^k = -B_k^{-1} g_k$. Assume that $g_k \neq 0$ for all $k \geq 0$. Since f is bounded below and ∇f is Lipschitz continuous, Theorem 2.9 gives

$$\lim_{k \rightarrow \infty} \min \left\{ \frac{|g_k^\top s^k|}{\|s^k\|}, |g_k^\top s^k| \right\} = 0. \quad (4.3)$$

It remains to show that $\|g_k\| \rightarrow 0$ as $k \rightarrow \infty$.

Since B_k is symmetric, B_k admits real eigenvalues by the spectral theorem. Since the eigenvalues are uniformly bounded above and away from zero, there exists constants $\lambda_{\max}$ and $\lambda_{\min}$ such that $\lambda_{\max} \geq \lambda_{\max}(B_k) \geq \lambda_{\min}(B_k) \geq \lambda_{\min}$ for all $k \geq 0$. In particular, for any $s \in \mathbb{R}^n$ and any B_k ,

$$\lambda_{\min} \|s\|^2 \leq \lambda_{\min}(B_k) \|s\|^2 \leq s^\top B_k s \leq \lambda_{\max}(B_k) \|s\|^2 \leq \lambda_{\max} \|s\|^2 \implies \lambda_{\min} \leq \frac{s^\top B_k s}{\|s\|^2} \leq \lambda_{\max}.$$

By definition of s^k and $g_k \neq 0$,

$$|g_k^\top s^k| = |g_k^\top B_k^{-1} g_k| \geq \lambda_{\min}(B_k^{-1}) \|g_k\|^2 \geq \frac{\|g_k\|^2}{\lambda_{\max}}.$$

On the other hand,

$$\|s^k\|^2 = g_k^\top B_k^{-2} g_k \leq \lambda_{\max}(B_k^{-2}) \|g_k\|^2 \leq \frac{\|g_k\|^2}{\lambda_{\min}^2} \implies \frac{1}{\|s^k\|} \geq \frac{\lambda_{\min}}{\|g_k\|}.$$

Plugging into (4.3), we have

$$\lim_{k \rightarrow \infty} \min \left\{ \frac{\lambda_{\min}}{\lambda_{\max}} \|g_k\|, \frac{\|g_k\|^2}{\lambda_{\max}} \right\} \leq \lim_{k \rightarrow \infty} \min \left\{ \frac{|g_k^\top s^k|}{\|s^k\|}, |g_k^\top s^k| \right\} \rightarrow 0 \implies \lim_{k \rightarrow \infty} g_k = 0.$$

$\square$

Corollary 4.9 (Newton with bArmijo: stepsize bound and global convergence). *Let $f \in \mathcal{C}^2(\mathbb{R}^n)$ be bounded below with ∇f L -Lipschitz continuous. Suppose the eigenvalues of $\nabla^2 f(x)$ satisfy $\lambda_{\max} \geq \lambda_{\max}(\nabla^2 f(x)) \geq \lambda_{\min}(\nabla^2 f(x)) \geq \lambda_{\min} > 0$ for all x . Apply Newton's method with bArmijo linesearch to f .*

1. *The bArmijo stepsize satisfies $\alpha_k \geq \min\{\alpha^{(0)}, \tau(1 - \beta)\lambda_{\min}^2/L^2\} =: C$ for all $k \geq 0$.*
2. *Newton with bArmijo is globally convergent: $\|\nabla f(x^k)\| \rightarrow 0$.*

Proof. The Newton direction $s^k = -[\nabla^2 f(x^k)]^{-1} \nabla f(x^k)$ is descent since $\nabla^2 f(x^k) \succ 0$. By Lemma 2.8, $\alpha_k \geq \min\{\alpha^{(0)}, \tau\alpha_{k,\max}\}$, where $\alpha_{k,\max} = (1 - \beta)|\nabla f(x^k)^\top s^k|/(L\|s^k\|^2)$. It remains to bound the ratio $|\nabla f(x^k)^\top s^k|/\|s^k\|^2$.

Write $g_k := \nabla f(x^k)$ and $H_k := \nabla^2 f(x^k)$. Then

$$|g_k^\top s^k| = g_k^\top H_k^{-1} g_k \geq \lambda_{\min}(H_k^{-1}) \|g_k\|^2 = \frac{\|g_k\|^2}{\lambda_{\max}(H_k)} \geq \frac{\|g_k\|^2}{L},$$

and

$$\|s^k\|^2 = g_k^\top H_k^{-2} g_k \leq \lambda_{\max}(H_k^{-2}) \|g_k\|^2 = \frac{\|g_k\|^2}{\lambda_{\min}(H_k)^2} \leq \frac{\|g_k\|^2}{\lambda_{\min}^2}.$$

Hence $\alpha_{k,\max} \geq (1 - \beta)\lambda_{\min}^2/L^2$, proving (1). For (2), the Armijo condition gives

$$f(x^k) - f(x^{k+1}) \geq \beta\alpha_k |g_k^\top s^k| \geq \frac{\beta C}{L} \|g_k\|^2.$$

Telescoping and using f bounded below: $\sum_{k=0}^{\infty} \|g_k\|^2 < \infty$, so $\|g_k\| \rightarrow 0$. □

4.5 SGD vs Newton

	Steepest Descent	Newton's Method
Search direction	$s^k = -\nabla f(x^k)$	$\nabla^2 f(x^k) s^k = -\nabla f(x^k)$
Info required	∇f only	Both ∇f and $\nabla^2 f$
Descent property	Always descent unless $\nabla f(x^k) = 0$ (then terminate)	$(s^k)^\top \nabla f(x^k) < 0$ iff $\nabla^2 f(x^k) \succ 0$
Local rate near x^*	Linear: $f(x^{k+1}) - f_* \leq \beta(f(x^k) - f_*)$, $\beta \in (0, 1)$; $\beta \approx 1$ when ill-conditioned	Quadratic: $f(x^{k+1}) - f_* \leq C(f(x^k) - f_*)^2$ — requires $\alpha_k = 1$ (or $\alpha_k \rightarrow 1$)
Scale invariance	No	Yes
Global convergence	f bounded below, ∇f Lipschitz $\implies \nabla f(x^k) \rightarrow 0$; f may be non-convex	Same plus $\nabla^2 f(x^k) \succeq \lambda_* I$ for some $\lambda_* > 0$ and all k (i.e. uniformly strongly convex) $\implies \nabla f(x^k) \rightarrow 0$
Main restriction	Slow when ill-conditioned	Needs $\nabla^2 f \succ 0$ globally — big restriction (essentially f convex)

Chapter 5

Quasi-Newton Methods and Nonlinear Least-Squares

Newton's method has fast local convergence but each iteration is expensive: forming $\nabla^2 f(x^k)$ costs $O(n^2)$ second-derivative evaluations, and solving the linear system costs $O(n^3)$. *Quasi-Newton methods* aim for a sweet spot: faster than steepest descent, but cheaper per iteration than Newton, by using *secant approximations* $B_k \approx \nabla^2 f(x^k)$ built from gradient information already computed.

The design principles for B_{k+1} form a “wish list”:

- built from already-computed quantities $\nabla f(x^{k+1}), \nabla f(x^k), \dots, B_k, s^k$;
- symmetric, nonsingular, and ideally positive definite;
- “close” to B_k and a “cheap” update of B_k ;
- asymptotically, $B_k \rightarrow \nabla^2 f(x^*)$.

No single update satisfies the entire list; different methods make different trade-offs.

Remark 5.1 (Approximating the Hessian by finite differences). An alternative when second derivatives are unavailable is to approximate $[\nabla^2 f(x)]e_i \approx \frac{1}{h}[\nabla f(x+he_i) - \nabla f(x)]$ for $i = 1, \dots, n$, costing $n + 1$ gradient evaluations. Choice of h is delicate (too large $\rightarrow$ inaccurate; too small $\rightarrow$ cancellation errors).

5.1 The secant equation

In quasi-Newton methods, $\nabla^2 f(x^{k+1})$ is approximated by a symmetric matrix B_{k+1} satisfying the *secant equation*

$$B_{k+1} \delta^k = \gamma^k, \quad \delta^k := x^{k+1} - x^k, \quad \gamma^k := \nabla f(x^{k+1}) - \nabla f(x^k). \quad (5.1)$$

Lemma 5.2 (Quadratic case). *If $f(x) = g^\top x + \frac{1}{2}x^\top Hx$, then (5.1) is satisfied with $B_{k+1} = \nabla^2 f = H$.*

Proof. Since $\nabla f(x) = Hx + g$,

$$\nabla f(x^{k+1}) - \nabla f(x^k) = H(x^{k+1} - x^k),$$

which satisfies the secant equation. □

5.2 The BFGS update

There are infinitely many ways to satisfy the secant equation. One choice is the *symmetric rank-1 (SR1)* update $B_{k+1} = B_k + u^k(u^k)^\top$, which is cheap ($O(n^2)$ per iteration via the Sherman–Morrison–Woodbury formula) but may fail to be positive definite, leading to non-descent directions. The *BFGS* update (Broyden, Fletcher, Goldfarb, Shanno, independently 1970) is a rank-2 update that preserves positive definiteness.

Definition 5.3 (BFGS update). The *BFGS* update is the rank-2 symmetric update

$$B_{k+1} = B_k - \frac{B_k \delta^k (\delta^k)^\top B_k}{(\delta^k)^\top B_k \delta^k} + \frac{\gamma^k (\gamma^k)^\top}{(\delta^k)^\top \gamma^k}. \quad (5.2)$$

Theorem 5.4 (Properties of the BFGS update). *If B_k is symmetric positive definite and $(\delta^k)^\top \gamma^k > 0$, then the BFGS update (5.2) is well-defined, symmetric and positive definite, and satisfies the secant equation (5.1).*

Proof. It is easy to see that B_{k+1} is symmetric, as the two terms after B_k are symmetric. For $x \in \mathbb{R}^n$ with $x \neq 0$,

$$x^\top B_{k+1} x = \underbrace{x^\top B_k x}_{=:P} - \frac{[(\delta^k)^\top B_k x]^2}{(\delta^k)^\top B_k \delta^k} + \underbrace{\frac{[(\gamma^k)^\top x]^2}{(\delta^k)^\top \gamma^k}}_{=:Q \geq 0}.$$

Since $B_k \succ 0$, we may define the inner product $\langle u, v \rangle_B = u^\top B_k v$. By Cauchy–Schwarz,

$$P = \langle x, x \rangle_B - \frac{\langle \delta^k, x \rangle_B^2}{\langle \delta^k, \delta^k \rangle_B} \geq 0,$$

so $x^\top B_{k+1} x \geq 0$. Suppose $x^\top B_{k+1} x = 0$. Then $P = 0$ and $Q = 0$. From $Q = 0$: $(\gamma^k)^\top x = 0$. From $P = 0$: Cauchy–Schwarz is tight, so $x = c \delta^k$ for some $c \in \mathbb{R}$ (in the B_k -inner product). But then $(\gamma^k)^\top x = c (\delta^k)^\top \gamma^k = 0$, and since $(\delta^k)^\top \gamma^k > 0$, we get $c = 0$, hence $x = 0$, a contradiction. Therefore $x^\top B_{k+1} x > 0$ for all $x \neq 0$.

Since

$$B_{k+1} \delta^k = B_k \delta^k - \frac{B_k \delta^k [(\delta^k)^\top B_k \delta^k]}{(\delta^k)^\top B_k \delta^k} + \frac{\gamma^k [(\gamma^k)^\top \delta^k]}{(\delta^k)^\top \gamma^k} = \gamma^k,$$

the secant equation is satisfied. $\square$

Lemma 5.5 (Wolfe conditions guarantee $(\delta^k)^\top \gamma^k > 0$). *If the stepsize α_k satisfies the Wolfe conditions (W1)–(W2) with $0 < \beta < \eta < 1$ and s^k is a descent direction, then $(\delta^k)^\top \gamma^k > 0$.*

Proof. Write $\delta^k = \alpha_k s^k$ and $\gamma^k = \nabla f(x^{k+1}) - \nabla f(x^k)$. The curvature condition (W2) gives

$$\nabla f(x^{k+1})^\top s^k \geq \eta \nabla f(x^k)^\top s^k.$$

Subtracting $\nabla f(x^k)^\top s^k$ from both sides,

$$(\gamma^k)^\top s^k \geq (\eta - 1) \nabla f(x^k)^\top s^k.$$

Since $\eta < 1$ and $\nabla f(x^k)^\top s^k < 0$, the right-hand side is strictly positive. Multiplying by $\alpha_k > 0$,

$$(\delta^k)^\top \gamma^k = \alpha_k (\gamma^k)^\top s^k \geq \alpha_k (\eta - 1) \nabla f(x^k)^\top s^k > 0. \quad \square$$

Theorem 5.6 (Convergence of BFGS). *Under suitable conditions on f and using the Wolfe linesearch, the BFGS method is globally convergent and locally Q -superlinearly convergent.* $\square$

Remark 5.7 (BFGS in practice). BFGS satisfies essentially all the items on the wish list: it is symmetric and positive definite (so $s^k = -B_k^{-1}\nabla f(x^k)$ is descent), close to B_k , cheap to update (in practice the Cholesky factors of B_k are stored and updated directly). Applied with exact linesearch to a strictly convex quadratic, B_k converges to $\nabla^2 f$ in at most n iterations. BFGS suffers much less from poor scaling than steepest descent. It has been the most popular general-purpose method when second derivatives are unavailable, and is the basis of MATLAB's `fminunc`.

5.3 Nonlinear least-squares

Many problems (model fitting, parameter estimation, data assimilation) take the form: choose x to minimise the sum of squared residuals $r_j(x)^2$. The problem is called *zero-residual* if $r(x^*) = 0$ (the model fits exactly) and *nonzero-residual* otherwise. The special structure of f allows significant savings over generic Newton methods.

The nonlinear least-squares problem is

$$\min_{x \in \mathbb{R}^n} f(x) = \frac{1}{2} \|r(x)\|^2 = \frac{1}{2} \sum_{j=1}^m r_j(x)^2, \quad (5.3)$$

where $r : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is smooth with Jacobian $J(x)$.

Lemma 5.8 (Derivatives of f). *For f in (5.3),*

$$\nabla f(x) = J(x)^\top r(x), \quad \nabla^2 f(x) = J(x)^\top J(x) + \sum_{j=1}^m r_j(x) \nabla^2 r_j(x).$$

Proof.

$$[\nabla f(x)]_i = \frac{\partial f}{\partial x_i}(x) = \sum_{j=1}^m r_j(x) \cdot \frac{\partial r_j}{\partial x_i}(x) = r(x)^\top \frac{\partial r}{\partial x_i}(x) = r(x)^\top J(x) e_i.$$

$$\begin{aligned} [\nabla^2 f(x)]_{ik} &= \frac{\partial^2 f}{\partial x_i \partial x_k}(x) \\ &= \sum_{j=1}^m \frac{\partial}{\partial x_k} \left[r_j(x) \cdot \frac{\partial r_j}{\partial x_i}(x) \right] \\ &= \sum_{j=1}^m \frac{\partial r_j}{\partial x_i}(x) \cdot \frac{\partial r_j}{\partial x_k}(x) + r_j(x) \cdot \frac{\partial^2 r_j}{\partial x_i \partial x_k}(x) \\ &= [J^\top(x) J(x)]_{ik} + \left[\sum_{j=1}^m r_j(x) \nabla^2 r_j(x) \right]_{ik}. \end{aligned}$$

$\square$

5.4 The Gauss-Newton method

Motivation. The Hessian decomposes as $\nabla^2 f(x) = J(x)^\top J(x) + \sum_j r_j(x) \nabla^2 r_j(x)$. When residuals are small ($r_j(x^*) \approx 0$) or the model is nearly linear ($\nabla^2 r_j \approx 0$), the second term is

negligible and $\nabla^2 f(x) \approx J(x)^\top J(x)$. The Gauss-Newton method substitutes this approximation into the Newton system, dispensing with second derivatives entirely.

Definition 5.9 (Gauss-Newton direction). The *Gauss-Newton (GN) direction* solves the linear system

$$J(x^k)^\top J(x^k) s^k = -J(x^k)^\top r(x^k). \quad (5.4)$$

Lemma 5.10 (Descent property). *If $J(x^k)$ has full column rank, then the Gauss-Newton direction is a descent direction for f .*

Proof. Since $J(x^k)$ has full column rank, $J(x^k)^\top J(x^k)$ is positive definite, and so $s^k \neq 0$ (as $\nabla f(x^k) = J(x^k)^\top r(x^k) \neq 0$). From (5.4),

$$[\nabla f(x^k)]^\top s^k = -[J(x^k)^\top J(x^k) s^k]^\top s^k = -(s^k)^\top J(x^k)^\top J(x^k) s^k < 0,$$

since $J(x^k)^\top J(x^k) \succ 0$ and $s^k \neq 0$. □

Theorem 5.11 (Convergence of Gauss-Newton). *Apply Gauss-Newton with bArmijo linesearch.*

1. (Global.) *If $J(x^k)$ is uniformly full rank, $f \in \mathcal{C}^1$ bounded below and ∇f Lipschitz, then $\|\nabla f(x^k)\| \rightarrow 0$.*
2. (Local.) *If $r(x^*) = 0$ and $J(x^*)$ has full rank, the convergence is quadratic; if $r(x^*) \neq 0$ but small, the convergence is linear.*

Part (1) follows from Theorem 4.8 with $B_k = J(x^k)^\top J(x^k)$ and the uniform full-rank assumption. We sketch the proof of the quadratic rate in part (2).

Proof sketch of local quadratic convergence (zero-residual case). Assume $r(x^*) = 0$, $J(x^*)$ full column rank, and J locally Lipschitz at x^* . Taylor-expand r around x^k (sufficiently close to x^*):

$$0 = r(x^*) = r(x^k) + J(x^k)(x^* - x^k) + O(\|x^* - x^k\|^2).$$

Since $J(x^k)$ has full column rank near x^* , the pseudo-inverse $J(x^k)^+ := (J(x^k)^\top J(x^k))^{-1} J(x^k)^\top$ is well-defined and bounded. Applying $J(x^k)^+$ and using $J(x^k)^+ J(x^k) = I$:

$$x^k - x^* = J(x^k)^+ r(x^k) + O(\|x^* - x^k\|^2).$$

The pure GN iterate (with $\alpha_k = 1$) is $x^{k+1} = x^k - J(x^k)^+ r(x^k)$, so

$$x^{k+1} - x^* = O(\|x^k - x^*\|^2),$$

giving quadratic convergence. □

Remark 5.12 (Newton vs. Gauss-Newton).

- **Cost per iteration:** Newton $>$ Gauss-Newton (no second derivatives needed for GN).
- **Newton direction may be ascent** if $\nabla^2 f(x^k)$ is not positive definite. In contrast, the GN direction is always descent provided $J(x^k)$ has full column rank.
- **Local rate:** Newton always quadratic; GN only quadratic for zero-residual problems and only linear when $r(x^*) \neq 0$.
- Both methods are unreliable without a linesearch (or trust-region) safeguard.

Chapter 6

Trust-Region Methods

Trust-region methods are an alternative to linesearch for globalising Newton-type iterations. The two philosophies differ:

- **Linesearch:** “liberal” in choice of search direction; keeps bad behaviour in control by choosing α_k .
- **Trust-region:** “conservative” in choice of step; the full step is accepted if a sufficient *model decrease* is realised in the objective. Otherwise the step is rejected and the model is refined.

A key advantage of TR methods is that they accommodate *indefinite* or ill-behaved Hessians: the trust-region constraint regularises an otherwise unbounded model. They also fit naturally with quasi-Newton ideas, since there is no need for B_k to be positive definite.

6.1 The trust-region framework

We approximate $f(x^k + s)$ by a local model. Two natural choices are:

- linear: $\ell_k(s) := f(x^k) + s^\top \nabla f(x^k)$;
- quadratic: $q_k(s) := f(x^k) + s^\top \nabla f(x^k) + \frac{1}{2} s^\top B_k s$, with $B_k = \nabla^2 f(x^k)$ in the standard case.

Impediments. The model may not resemble f when $\|s\|$ is large, and may be unbounded below: ℓ_k is always unbounded below (unless $\nabla f(x^k) = 0$), and q_k is unbounded below whenever B_k is negative definite or indefinite, and sometimes when B_k is only positive semidefinite. We trust the model only on a region $\|s\| \leq \Delta_k$: At iteration k , we form a quadratic model

$$m_k(s) = f(x^k) + s^\top \nabla f(x^k) + \frac{1}{2} s^\top B_k s, \quad (6.1)$$

where B_k is symmetric (e.g. $B_k = \nabla^2 f(x^k)$). The *trust-region (TR) subproblem* is

$$\min_{s \in \mathbb{R}^n} m_k(s) \quad \text{subject to} \quad \|s\| \leq \Delta_k, \quad (\text{TR})$$

with trust-region radius $\Delta_k > 0$.

Algorithm 6.1 (Generic Trust-Region (GTR) method). Given $\Delta_0 > 0$, $x^0 \in \mathbb{R}^n$, $\varepsilon > 0$. While $\|\nabla f(x^k)\| > \varepsilon$:

1. Form m_k and (approximately) solve (TR) for s^k with $m_k(s^k) < f(x^k)$.
2. Compute $\rho_k = \frac{f(x^k) - f(x^k + s^k)}{f(x^k) - m_k(s^k)}$.
3. If $\rho_k \geq 0.9$: set $x^{k+1} := x^k + s^k$, $\Delta_{k+1} := 2\Delta_k$.
 Else if $\rho_k \geq 0.1$: set $x^{k+1} := x^k + s^k$, $\Delta_{k+1} := \Delta_k$.
 Else: set $x^{k+1} := x^k$, $\Delta_{k+1} := \frac{1}{2}\Delta_k$.

Remark 6.2 (The ratio ρ_k). ρ_k measures the ratio of the *actual function decrease* to the *predicted model decrease*. $\rho_k \approx 1$ indicates the model is accurate (we expand the trust region to take longer steps); $\rho_k \ll 1$ indicates the model is poor (we reject the step and shrink the trust region). The thresholds 0.1, 0.9 are conventional; other reasonable values appear in practice.

6.2 The Cauchy point

The Cauchy point is the trust-region analogue of the steepest descent direction: it minimises the model m_k along the steepest descent direction subject to the trust-region constraint. Recall (Theorem 3.3) that steepest descent has strong global convergence properties; insisting that any approximate solution s^k of the TR subproblem do at least as well as the Cauchy point is a “minimal sufficient decrease” condition that suffices for global convergence of GTR.

Definition 6.3 (Cauchy point). The *Cauchy point* of (TR) is $y_c^k = x^k + s_c^k$, where $s_c^k = -\alpha_{k,c} \nabla f(x^k)$ and

$$\alpha_{k,c} = \arg \min_{\alpha > 0} m_k(-\alpha \nabla f(x^k)) \quad \text{subject to} \quad \|\alpha \nabla f(x^k)\| \leq \Delta_k.$$

Lemma 6.4 (Cauchy-point decrease).

$$f(x^k) - m_k(s_c^k) \geq \frac{1}{2} \|\nabla f(x^k)\| \min \left\{ \Delta_k, \frac{\|\nabla f(x^k)\|}{\|B_k\|} \right\}.$$

Proof. Write $g_k := \nabla f(x^k)$ and $h_k = \nabla f(x^k)^\top B_k \nabla f(x^k)$. Define

$$\psi(\alpha) := m_k(-\alpha g_k) = f(x^k) - \alpha \|g_k\|^2 + \frac{1}{2} \alpha^2 h_k, \quad \alpha > 0,$$

so that the Cauchy step is $s_c^k = -\alpha_c^k g_k$ with

$$\alpha_c^k = \arg \min_{0 < \alpha \leq \bar{\alpha}} \psi(\alpha), \quad \bar{\alpha} := \frac{\Delta_k}{\|g_k\|}.$$

Note $\psi'(0) = -\|g_k\|^2 < 0$, so ψ is initially decreasing and $\alpha_c^k > 0$.

Case 1: $h_k \leq 0$. Then ψ is concave (or linear) and strictly decreasing on $(0, \bar{\alpha}]$, so the minimum on the trust region is attained at the boundary: $\alpha_c^k = \bar{\alpha} = \Delta_k / \|g_k\|$. Since the quadratic term $\frac{1}{2} \alpha^2 h_k \leq 0$,

$$f(x^k) - m_k(s_c^k) = \alpha_c^k \|g_k\|^2 - \frac{1}{2} (\alpha_c^k)^2 h_k \geq \alpha_c^k \|g_k\|^2 = \Delta_k \|g_k\|.$$

Case 2: $h_k > 0$. Now ψ is a strictly convex quadratic in α with unconstrained minimiser

$$\alpha_{\min} = \frac{\|g_k\|^2}{h_k},$$

and $\alpha_c^k = \min\{\alpha_{\min}, \bar{\alpha}\}$. We consider the two sub-cases.

Sub-case 2a: $\alpha_{\min} \leq \bar{\alpha}$. Then $\alpha_c^k = \alpha_{\min}$ and substituting gives

$$f(x^k) - m_k(s_c^k) = \alpha_{\min} \|g_k\|^2 - \frac{1}{2} \alpha_{\min}^2 h_k = \frac{\|g_k\|^4}{2h_k}.$$

By Cauchy–Schwarz and the Rayleigh-quotient bound, $h_k = g_k^\top B_k g_k \leq \|B_k\| \|g_k\|^2$, hence

$$f(x^k) - m_k(s_c^k) \geq \frac{\|g_k\|^2}{2\|B_k\|} \geq \frac{1}{2} \|g_k\| \min \left\{ \Delta_k, \frac{\|g_k\|}{\|B_k\|} \right\}.$$

Sub-case 2b: $\alpha_{\min} > \bar{\alpha}$. Then $\alpha_c^k = \bar{\alpha} = \Delta_k / \|g_k\|$. Since ψ is decreasing on $[0, \alpha_{\min})$, evaluating at $\bar{\alpha}$ and using $\bar{\alpha} < \alpha_{\min} = \|g_k\|^2 / h_k$, i.e. $\bar{\alpha} h_k < \|g_k\|^2$,

$$f(x^k) - m_k(s_c^k) = \bar{\alpha} \|g_k\|^2 - \frac{1}{2} \bar{\alpha}^2 h_k > \bar{\alpha} \|g_k\|^2 - \frac{1}{2} \bar{\alpha} \|g_k\|^2 = \frac{1}{2} \bar{\alpha} \|g_k\|^2 = \frac{1}{2} \Delta_k \|g_k\|.$$

This completes the proof. $\square$

Remark 6.5. Lemma 6.4 is the key technical estimate underlying global convergence: any step that does at least as well as the Cauchy step inherits this lower bound on the model decrease, and combined with the ratio test in GTR this drives $\|\nabla f(x^k)\| \rightarrow 0$.

6.3 Global convergence of GTR

Theorem 6.6 (Global convergence of GTR). *Let $f \in \mathcal{C}^2(\mathbb{R}^n)$ be bounded below with ∇f L -Lipschitz continuous, and assume $\|B_k\|$ is uniformly bounded. Apply GTR (Algorithm 6.1) requiring $m_k(s^k) \leq m_k(s_c^k)$. Then either there exists $\ell \geq 0$ with $\nabla f(x^\ell) = 0$, or $\liminf_{k \rightarrow \infty} \|\nabla f(x^k)\| = 0$.*

The proof rests on two lemmas: the Cauchy-point decrease (Lemma 6.4) and the lower bound on the trust-region radius below.

Lemma 6.7 (Lower bound on the TR radius). *Under the assumptions of Theorem 6.6, suppose there exists $\varepsilon > 0$ such that $\|\nabla f(x^k)\| \geq \varepsilon$ for all k . Then there exists a constant $c \in (0, 1)$ such that*

$$\Delta_k \geq \frac{c\varepsilon}{L} \quad \text{for all } k \geq 0.$$

Proof. Write $g_k := \nabla f(x^k)$. By the integral form of Taylor’s theorem and $\|\nabla^2 f(x)\| \leq L$,

$$|f(x^k + s^k) - m_k(s^k)| \leq \frac{L}{2} \|s^k\|^2 \leq \frac{L}{2} \Delta_k^2.$$

Combined with the Cauchy-point bound (Lemma 6.4) and $\|B_k\| = \|\nabla^2 f(x^k)\| \leq L$,

$$f(x^k) - m_k(s^k) \geq \frac{1}{2} \varepsilon \min \left\{ \Delta_k, \frac{\varepsilon}{L} \right\}.$$

Hence

$$|\rho_k - 1| = \frac{|f(x^k + s^k) - m_k(s^k)|}{f(x^k) - m_k(s^k)} \leq \frac{(L/2) \Delta_k^2}{(\varepsilon/2) \min\{\Delta_k, \varepsilon/L\}}.$$

Now choose $c := \min \left\{ \frac{1}{2}, \frac{\Delta_0 L}{\varepsilon} \right\} \in (0, 1)$. Suppose, for contradiction, that some iteration k has $\Delta_k \leq 2c\varepsilon/L \leq \varepsilon/L$. Then $\min\{\Delta_k, \varepsilon/L\} = \Delta_k$, and substituting,

$$|\rho_k - 1| \leq \frac{(L/2) \Delta_k^2}{(\varepsilon/2) \Delta_k} = \frac{L \Delta_k}{\varepsilon} \leq 2c \leq 1 - 0.9 + c \leq 0.1.$$

Therefore $\rho_k \geq 0.9$, so iteration k is *very successful* and the GTR rule sets $\Delta_{k+1} = 2\Delta_k \geq \Delta_k$.

Thus the radius can never drop below $c\varepsilon/L$: any iteration with $\Delta_k \leq 2c\varepsilon/L$ at least preserves the radius, so by induction on k ,

$$\Delta_k \geq \min\{\Delta_0, \frac{c\varepsilon}{L}\} \geq \frac{c\varepsilon}{L}$$

for all $k \geq 0$ (after possibly shrinking c further so that $\Delta_0 \geq c\varepsilon/L$). This proves the claim. $\square$

Proof of Theorem 6.6. If $\nabla f(x^k) = 0$ for some k , the algorithm terminates and there is nothing to prove; this also covers the case of finitely many successful iterations, since on every unsuccessful step $\Delta_{k+1} = \frac{1}{2}\Delta_k$, so $\Delta_k \rightarrow 0$, contradicting Lemma 6.7 unless the gradient at the last successful iterate is zero.

Suppose instead, for contradiction, that there exists $\varepsilon > 0$ with $\|\nabla f(x^k)\| \geq \varepsilon$ for all k , and that the set $\mathcal{S} := \{k : \rho_k \geq 0.1\}$ of successful iterations is infinite. For every $k \in \mathcal{S}$ the GTR update gives $x^{k+1} = x^k + s^k$ and

$$f(x^k) - f(x^{k+1}) \geq 0.1(f(x^k) - m_k(s^k)) \geq \frac{0.1}{2} \|g_k\| \min\left\{\Delta_k, \frac{\|g_k\|}{\|B_k\|}\right\},$$

by Lemma 6.4. Since $\|g_k\| \geq \varepsilon$ and $\|B_k\| = \|\nabla^2 f(x^k)\| \leq L$,

$$f(x^k) - f(x^{k+1}) \geq 0.05\varepsilon \min\left\{\Delta_k, \frac{\varepsilon}{L}\right\} \geq 0.05\varepsilon \min\left\{\frac{c\varepsilon}{L}, \frac{\varepsilon}{L}\right\} = \frac{0.05c}{L} \varepsilon^2, \quad (*)$$

where the second inequality uses Lemma 6.7 and $c \in (0, 1)$.

Since $f(x^{k+1}) = f(x^k)$ on every unsuccessful iteration, telescoping the decrease over all $k \leq K-1$ gives

$$f(x^0) - f(x^K) = \sum_{i=0}^{K-1} (f(x^i) - f(x^{i+1})) = \sum_{i \in \mathcal{S}, i < K} (f(x^i) - f(x^{i+1})).$$

Letting $K \rightarrow \infty$ and using $f(x^K) \geq f_{\text{low}}$,

$$f(x^0) - f_{\text{low}} \geq \sum_{i \in \mathcal{S}} (f(x^i) - f(x^{i+1})) \geq |\mathcal{S}| \cdot \frac{0.05c}{L} \varepsilon^2 \rightarrow \infty, \quad (**)$$

by (*). But then left-hand side is finite by assumption, contradiction. Hence no such $\varepsilon > 0$ exists, and so $\liminf_{k \rightarrow \infty} \|\nabla f(x^k)\| = 0$. $\square$

Theorem 6.8 (Stronger global convergence). *Under the conditions of Theorem 6.6,*

$$\lim_{k \rightarrow \infty} \|\nabla f(x^k)\| = 0.$$

$\square$

6.4 Solving the trust-region subproblem

The characterisation below shows that solving (TR) reduces to the secular equation

$$\|(B_k + \lambda I)^{-1} \nabla f(x^k)\| = \Delta_k$$

in the scalar $\lambda \geq \max\{0, -\lambda_{\min}(B_k)\}$.

Theorem 6.9 (Characterization of the global TR solution). *A vector $s^* \in \mathbb{R}^n$ is a global solution of (TR) if and only if $\|s^*\| \leq \Delta_k$ and there exists $\lambda^* \geq 0$ such that*

1. $(B_k + \lambda^* I)s^* = -\nabla f(x^k)$,
2. $\lambda^*(\Delta_k - \|s^*\|) = 0$ (complementarity),
3. $B_k + \lambda^* I \succeq 0$.

Moreover λ^* is unique whenever $B_k + \lambda^* I \succ 0$.

We prove the sufficiency direction (the necessity direction uses Lagrangian duality and is not given here).

Proof of sufficiency. Write $g := \nabla f(x^k)$ and set $B := B_k + \lambda^* I \succeq 0$. Condition (1) reads $Bs^* = -g$, which says s^* is a stationary point of the unconstrained quadratic $M(s) := s^\top g + \frac{1}{2}s^\top Bs$. Since $B \succeq 0$, M is convex and s^* is its global minimizer: $M(s) \geq M(s^*)$ for all $s \in \mathbb{R}^n$.

Now the original model is $m_k(s) = f(x^k) + s^\top g + \frac{1}{2}s^\top B_k s = f(x^k) + M(s) - \frac{\lambda^*}{2}\|s\|^2$. For any feasible s with $\|s\| \leq \Delta_k$,

$$m_k(s) = f(x^k) + M(s) - \frac{\lambda^*}{2}\|s\|^2 \geq f(x^k) + M(s^*) - \frac{\lambda^*}{2}\|s\|^2 = m_k(s^*) + \frac{\lambda^*}{2}(\|s^*\|^2 - \|s\|^2).$$

By complementarity (2), $\lambda^*(\Delta_k - \|s^*\|) = 0$, so $\lambda^*\|s^*\|^2 = \lambda^*\Delta_k^2$ (either $\lambda^* = 0$ or $\|s^*\| = \Delta_k$). Since $\|s\| \leq \Delta_k$ and $\lambda^* \geq 0$,

$$\frac{\lambda^*}{2}(\|s^*\|^2 - \|s\|^2) = \frac{\lambda^*}{2}(\Delta_k^2 - \|s\|^2) \geq 0.$$

Hence $m_k(s) \geq m_k(s^*)$ for every feasible s . □

Theorem 6.9 gives necessary and sufficient global optimality conditions for a nonconvex optimization problem. It splits the practical solution into two cases:

- **Case 1:** $B_k \succ 0$ and the Newton step $s = -B_k^{-1}\nabla f(x^k)$ satisfies $\|s\| \leq \Delta_k$ — then $s^* = s$ and $\lambda^* = 0$ (Newton step is inside the trust region).
- **Case 2:** otherwise — find $\lambda > \max\{0, -\lambda_{\min}(B_k)\}$ with $\|(B_k + \lambda I)^{-1}\nabla f(x^k)\| = \Delta_k$, by a 1-D root-find; the solution s^* lies on the boundary.

Remark 6.10 (Large-scale problems). For large n , repeated Cholesky factorisations of $B_k + \lambda I$ become prohibitive. In practice one uses iterative methods (truncated conjugate gradient, Lanczos / Steihaug–Toint) that produce an approximate solution at least as good as the Cauchy point, hence preserving the global convergence guarantee.

Remark 6.11 (Linesearch vs. trust-region in practice). Both linesearch and trust-region methods underpin state-of-the-art unconstrained optimisation software. Linesearch needs more heuristics to deal with negative curvature; TR handles indefinite Hessians naturally and accommodates quasi-Newton B_k that need not be positive definite. The choice between them is largely a matter of taste.

Chapter 7

Optimality Conditions for Constrained Optimization

The unconstrained theory characterised optimality by $\nabla f(x^*) = 0$. For constrained problems this fails: the gradient at the minimiser need not vanish, because feasibility prevents further descent. The right generalisation is the *Karush–Kuhn–Tucker (KKT)* conditions, which characterise points at which no *feasible* descent direction exists.

The development proceeds in three steps:

1. Define the geometric (tangent cone T_Ω) and algebraic (linearized feasible directions $\mathcal{F}$) descriptions of feasible directions; always $T_\Omega \subseteq \mathcal{F}$, but the reverse can fail.
2. Identify a *constraint qualification (CQ)* ensuring $T_\Omega = \mathcal{F}$ — without a CQ, optimality conditions need not hold.
3. Derive the KKT conditions from the algebraic description $\mathcal{F}$.

7.1 The constrained problem

We consider the general constrained problem

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad c_i(x) = 0, \ i \in E; \ c_i(x) \geq 0, \ i \in I, \quad (\text{CP})$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $c_E : \mathbb{R}^n \rightarrow \mathbb{R}^m$ collects the equality constraints $(c_i)_{i \in E}$, and $c_I : \mathbb{R}^n \rightarrow \mathbb{R}^p$ collects the inequality constraints $(c_i)_{i \in I}$. The condition $c_I(x) \geq 0$ is componentwise. The *feasible set* is

$$\Omega = \{x \in \mathbb{R}^n : c_E(x) = 0, \ c_I(x) \geq 0\},$$

and the *active set* at $x \in \Omega$ is $\mathcal{A}(x) = E \cup \{i \in I : c_i(x) = 0\}$.

7.2 Tangent cone and linearized feasible directions

Definition 7.1 (Tangent cone). The *tangent cone* to Ω at $x \in \Omega$ is the set of limits of feasible difference quotients,

$$T_\Omega(x) = \left\{ s \in \mathbb{R}^n : \exists z^k \in \Omega, \ t^k \downarrow 0, \ z^k \rightarrow x, \ s = \lim_{k \rightarrow \infty} \frac{z^k - x}{t^k} \right\}.$$

This is the *geometric* description of feasible directions.

Definition 7.2 (Linearized feasible directions). The set of *linearized feasible directions* at $x \in \Omega$ is

$$\mathcal{F}(x) = \{s \in \mathbb{R}^n : \nabla c_i(x)^\top s = 0, i \in E; \nabla c_i(x)^\top s \geq 0, i \in I \cap \mathcal{A}(x)\}.$$

This is the *algebraic* description of feasible directions.

Lemma 7.3. $T_\Omega(x) \subseteq \mathcal{F}(x)$.

Proof. Let $s \in T_\Omega(x)$, so there exist $z^k \in \Omega$ and $t^k \downarrow 0$ with $(z^k - x)/t^k \rightarrow s$. For $i \in E$, $c_i(z^k) = c_i(x) = 0$, so by Taylor,

$$0 = c_i(z^k) - c_i(x) = \nabla c_i(x)^\top (z^k - x) + o(\|z^k - x\|).$$

Dividing by t^k and letting $k \rightarrow \infty$ gives $\nabla c_i(x)^\top s = 0$. For $i \in I \cap \mathcal{A}(x)$, $c_i(z^k) \geq 0$ and $c_i(x) = 0$, so the same expansion yields $\nabla c_i(x)^\top s \geq 0$. $\square$

The reverse inclusion $\mathcal{F}(x) \subseteq T_\Omega(x)$ can fail: a direction in $\mathcal{F}$ that is not realisable by any feasible sequence is “algebraically feasible but geometrically not”.

Example 7.4 ($T_\Omega \subsetneq \mathcal{F}$). Let $\Omega = \{x \in \mathbb{R}^2 : c_1(x) \geq 0, c_2(x) \geq 0\}$ with

$$c_1(x) = 1 - x_1^2 - (x_2 - 1)^2, \quad c_2(x) = 1 - x_1^2 - (x_2 + 1)^2.$$

The two unit disks centred at $(0, 1)$ and $(0, -1)$ touch only at the origin, so $\Omega = \{(0, 0)\}$ and $T_\Omega(0) = \{(0, 0)\}$. On the other hand, $\nabla c_1(0) = (0, 2)^\top$ and $\nabla c_2(0) = (0, -2)^\top$, so

$$\mathcal{F}(0) = \{s \in \mathbb{R}^2 : s_2 \geq 0, -s_2 \geq 0\} = \{(s_1, 0) : s_1 \in \mathbb{R}\}.$$

Thus $T_\Omega(0) \subsetneq \mathcal{F}(0)$.

7.3 Constraint qualifications

We need a regularity condition on the constraints that rules out pathologies like Example 7.4.

Definition 7.5 (LICQ). (CP) satisfies the *Linear Independence Constraint Qualification (LICQ)* at $x \in \Omega$ if the active constraint gradients $\{\nabla c_i(x) : i \in \mathcal{A}(x)\}$ are linearly independent.

Definition 7.6 (Slater CQ). A convex problem (CP) (with $c_E(x) = Ax - b$ affine and c_i concave for $i \in I$) satisfies the *Slater CQ* if there exists $\bar{x}$ with $A\bar{x} = b$ and $c_i(\bar{x}) > 0$ for all $i \in I$.

Theorem 7.7 (CQ implies $T_\Omega = \mathcal{F}$). Under LICQ (or Slater, in the convex case), $T_\Omega(x) = \mathcal{F}(x)$. $\square$

In Example 7.4, $\nabla c_1(0)$ and $\nabla c_2(0)$ are parallel, so LICQ fails; the Slater condition also fails because no point satisfies both $c_1 > 0$ and $c_2 > 0$.

7.4 First-order necessary conditions (KKT)

Definition 7.8 (Lagrangian). The *Lagrangian* of (CP) is

$$\mathcal{L}(x, y, \lambda) = f(x) - \sum_{i \in E} y_i c_i(x) - \sum_{i \in I} \lambda_i c_i(x) = f(x) - y^\top c_E(x) - \lambda^\top c_I(x).$$

Definition 7.9 (KKT point). A feasible point $\hat{x} \in \Omega$ is a *KKT point* of (CP) if there exist $\hat{y} \in \mathbb{R}^m$ and $\hat{\lambda} \in \mathbb{R}^p$ such that

$$\begin{aligned}\nabla f(\hat{x}) &= \sum_{i \in E} \hat{y}_i \nabla c_i(\hat{x}) + \sum_{i \in I} \hat{\lambda}_i \nabla c_i(\hat{x}), \\ c_E(\hat{x}) &= 0, \quad c_I(\hat{x}) \geq 0, \\ \hat{\lambda}_i &\geq 0, \quad \hat{\lambda}_i c_i(\hat{x}) = 0, \quad i \in I.\end{aligned}$$

The last line is the *complementarity* condition. Equivalently, $\nabla_x \mathcal{L}(\hat{x}, \hat{y}, \hat{\lambda}) = 0$ together with primal feasibility, dual feasibility ($\hat{\lambda} \geq 0$), and complementarity.

The vectors $\hat{y}$ and $\hat{\lambda}$ are called the *Lagrange multipliers* for the equality and inequality constraints respectively. They arise naturally in sensitivity analysis: $\hat{\lambda}_i$ measures the sensitivity of the optimal value to a perturbation of $c_i \geq 0$.

Theorem 7.10 (KKT first-order necessary conditions). *Let x^* be a local minimizer of (CP) at which a CQ holds. Then x^* is a KKT point of (CP).*

We give the proof of Theorem 7.10 for the equality-constrained case ($I = \emptyset$).

Proof for equality constraints only. The KKT conditions reduce to $c_E(x^*) = 0$ (which holds by feasibility) and

$$\nabla f(x^*) = J_E(x^*)^\top y^* \quad \text{for some } y^* \in \mathbb{R}^m, \quad (7.1)$$

where J_E is the Jacobian of c_E .

Step 1: feasible directions satisfy $J_E(x^)s = 0$.* Consider a feasible $\mathcal{C}^1$ path $x : (-\delta, \delta) \rightarrow \Omega$ with $x(0) = x^*$ and $x(\alpha) = x^* + \alpha s + O(\alpha^2)$, $s \neq 0$, so that $c_E(x(\alpha)) = 0$ for all $\alpha \in (-\delta, \delta)$. Existence of such a path tangent to a given s requires the CQ. For each $i \in E$, Taylor's theorem gives

$$0 = c_i(x(\alpha)) = c_i(x^*) + \nabla c_i(x^*)^\top (\alpha s) + O(\alpha^2) = \alpha \nabla c_i(x^*)^\top s + O(\alpha^2),$$

using $c_i(x^*) = 0$. Dividing by α and letting $\alpha \rightarrow 0$ yields $\nabla c_i(x^*)^\top s = 0$ for all $i \in E$, i.e. $J_E(x^*)s = 0$.

Step 2: $\nabla f(x^)^\top s = 0$ for every such s .* Since x^* is a local minimizer and $x(\alpha) \in \Omega$ for α small, $f(x(\alpha)) \geq f(x^*)$ for all $\alpha \in (-\delta', \delta')$, some $\delta' > 0$. Taylor's theorem gives

$$0 \leq f(x(\alpha)) - f(x^*) = \alpha \nabla f(x^*)^\top s + O(\alpha^2).$$

Dividing by $\alpha > 0$ and letting $\alpha \downarrow 0$ gives $\nabla f(x^*)^\top s \geq 0$. Dividing by $\alpha < 0$ and letting $\alpha \uparrow 0$ gives $\nabla f(x^*)^\top s \leq 0$. Hence $\nabla f(x^*)^\top s = 0$ for every $s \in \ker J_E(x^*)$.

Step 3: $\nabla f(x^) \in \text{range } J_E(x^*)^\top$.* By the rank-nullity theorem, $\mathbb{R}^n = \text{range } J_E(x^*)^\top \oplus \ker J_E(x^*)$, so we can write

$$\nabla f(x^*) = J_E(x^*)^\top y^* + s^*, \quad J_E(x^*)s^* = 0, \quad (7.2)$$

for some $y^* \in \mathbb{R}^m$. Taking the inner product of (7.2) with s^* and using $J_E(x^*)s^* = 0$,

$$\nabla f(x^*)^\top s^* = (J_E(x^*)^\top y^*)^\top s^* + \|s^*\|^2 = y^{*\top} J_E(x^*)s^* + \|s^*\|^2 = \|s^*\|^2.$$

But $s^* \in \ker J_E(x^*)$, so by Step 2 the left-hand side is zero. Hence $s^* = 0$, and (7.2) reduces to (7.1). $\square$

Remark 7.11 (Where does the CQ enter?). The proof of Theorem 7.10 uses a feasible perturbation tangent to a prescribed $s \in \ker J_E(x^*)$. The CQ guarantees the existence of such a path: under LICQ, the implicit function theorem reduces the equality constraints to a smooth manifold whose tangent space is precisely $\ker J_E(x^*)$. Without a CQ, $T_\Omega \subsetneq \mathcal{F}$ is possible (Example 7.4), and the necessary conditions can fail.

Theorem 7.12 (Linear constraints need no CQ). *Suppose $(c_E, c_I)(x) = Ax - b$ is affine, with $A = \begin{pmatrix} A_E \\ A_I \end{pmatrix}$ and $b = \begin{pmatrix} b_E \\ b_I \end{pmatrix}$. Then any local minimizer x^* of (CP) is a KKT point: there exist $y^* \in \mathbb{R}^m$ and $\lambda^* \in \mathbb{R}^p$ with*

$$\begin{aligned} \nabla f(x^*) &= A_E^\top y^* + A_I^\top \lambda^*, \\ A_E x^* &= b_E, \quad A_I x^* \geq b_I, \\ \lambda^* &\geq 0, \quad (\lambda^*)^\top (A_I x^* - b_I) = 0. \end{aligned}$$

□

7.5 Sufficiency for convex problems

(CP) is a *convex programming problem* if Ω is a convex set and f is convex on Ω .

Theorem 7.13. (CP) is a convex programming problem if and only if f is convex over Ω and

- $c_E(x) = Ax - b = 0$
- $c_i(x)$ is concave for all $i \in I$,

Proof. Let $x, y \in \Omega$ and put $z = \alpha x + (1 - \alpha)y$ for $\alpha \in [0, 1]$. Then,

$$c_E(z) = Az - b = \alpha(Ax - b) + (1 - \alpha)(Ay - b) = 0$$

and for $i \in I$, by concavity of c_i ,

$$c_i(z) \geq \alpha c_i(x) + (1 - \alpha)c_i(y) \geq 0.$$

Hence, $z \in \Omega$.

□

For such problems, every local minimizer is global, and the KKT conditions are not just necessary (under a CQ) but also sufficient.

Theorem 7.14 (KKT is sufficient for convex problems). *Let (CP) be a convex programming problem and let $\hat{x}$ be a KKT point. Then $\hat{x}$ is a global minimizer of (CP).*

Proof. By convexity of f (first-order characterization),

$$f(x) \geq f(\hat{x}) + \nabla f(\hat{x})^\top (x - \hat{x}) \quad \text{for all } x \in \mathbb{R}^n. \quad (7.3)$$

The KKT stationarity condition reads $\nabla f(\hat{x}) = A^\top \hat{y} + \sum_{i \in I} \hat{\lambda}_i \nabla c_i(\hat{x})$. Substituting into (7.3),

$$f(x) \geq f(\hat{x}) + \hat{y}^\top A(x - \hat{x}) + \sum_{i \in I} \hat{\lambda}_i \nabla c_i(\hat{x})^\top (x - \hat{x}). \quad (7.4)$$

Now fix any $x \in \Omega$. Then $Ax = b = A\hat{x}$, so $A(x - \hat{x}) = 0$, killing the equality-multiplier term in (7.4).

For each $i \in I$, c_i is concave, hence $c_i(x) \leq c_i(\hat{x}) + \nabla c_i(\hat{x})^\top (x - \hat{x})$, giving

$$\nabla c_i(\hat{x})^\top (x - \hat{x}) \geq c_i(x) - c_i(\hat{x}).$$

Multiplying by $\hat{\lambda}_i \geq 0$ and using complementarity $\hat{\lambda}_i c_i(\hat{x}) = 0$,

$$\hat{\lambda}_i \nabla c_i(\hat{x})^\top (x - \hat{x}) \geq \hat{\lambda}_i c_i(x) - \hat{\lambda}_i c_i(\hat{x}) = \hat{\lambda}_i c_i(x) \geq 0,$$

the last inequality because $\hat{\lambda}_i \geq 0$ and $c_i(x) \geq 0$. Substituting back into (7.4) gives $f(x) \geq f(\hat{x})$. $\square$

7.6 Second-order conditions

The second-order theory mirrors the unconstrained case, but only on a restricted set of directions where the first-order information is inconclusive.

Definition 7.15 (Critical cone). At a KKT point x^* with multipliers (y^*, λ^*) , the *critical cone* is

$$\mathcal{F}(\lambda^*) = \{s \in \mathcal{F}(x^*) : \nabla c_i(x^*)^\top s = 0 \text{ for all } i \in I \cap \mathcal{A}(x^*) \text{ with } \lambda_i^* > 0\}.$$

The critical cone discards directions where the first-order term $\nabla f(x^*)^\top s$ is strictly positive (so f already increases at first order) and retains only those where it is zero. By the KKT identity, for $s \in \mathcal{F}(x^*)$,

$$\nabla f(x^*)^\top s = \underbrace{(J_E(x^*)s)^\top y^*}_{=0} + \sum_{i \in I \cap \mathcal{A}(x^*)} \lambda_i^* \nabla c_i(x^*)^\top s \geq 0,$$

with equality iff $\nabla c_i(x^*)^\top s = 0$ whenever $\lambda_i^* > 0$ — this is exactly $\mathcal{F}(\lambda^*)$.

Theorem 7.16 (Second-order necessary conditions). *Let x^* be a local minimizer of (CP) at which a CQ holds, with multipliers (y^*, λ^*) . Then $s^\top \nabla_{xx}^2 \mathcal{L}(x^*, y^*, \lambda^*) s \geq 0$ for all $s \in \mathcal{F}(\lambda^*)$. $\square$*

Theorem 7.17 (Second-order sufficient conditions). *Let x^* be feasible for (CP), and let (y^*, λ^*) satisfy the KKT conditions at x^* . If $s^\top \nabla_{xx}^2 \mathcal{L}(x^*, y^*, \lambda^*) s > 0$ for all $0 \neq s \in \mathcal{F}(\lambda^*)$, then x^* is a strict local minimizer of (CP). $\square$*

Here

$$\nabla_{xx}^2 \mathcal{L}(x, y, \lambda) = \nabla^2 f(x) - \sum_{j \in E} y_j \nabla^2 c_j(x) - \sum_{i \in I} \lambda_i \nabla^2 c_i(x).$$

Note that the second-order conditions involve the Hessian of the Lagrangian rather than of f alone — the constraint curvature, weighted by the multipliers, plays a genuine role.

Chapter 8

Penalty Methods

Constrained optimisation introduces a fundamental tension: we want to minimise f *and* satisfy the constraints. For inequality-constrained or linear equality-constrained problems we can sometimes maintain feasibility throughout, but with general nonlinear equality constraints this is very hard, even impossible. The idea behind *penalty methods* is to replace the constrained problem by a sequence of unconstrained problems whose minimisers approach the original solution as a parameter is varied.

We now turn to algorithms for the equality-constrained problem

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad c(x) = 0, \quad (\text{eCP})$$

where $c = (c_1, \dots, c_m) : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is smooth.

8.1 The quadratic penalty function

Definition 8.1 (Quadratic penalty function). For $\sigma > 0$, the *quadratic penalty function* associated with (eCP) is

$$\Phi_\sigma(x) = f(x) + \frac{1}{2\sigma} \|c(x)\|^2. \quad (8.1)$$

Interpretation. The parameter $\sigma > 0$ controls the penalty on infeasibility. As $\sigma \rightarrow 0$, infeasibility is forced down and the unconstrained minimisers of Φ_σ approach a solution of (eCP). The function Φ_σ may have stationary points that are *not* solutions of (eCP) — for example, local minimisers of $\|c(x)\|$ when $c(x) = 0$ has no solution.

Algorithm 8.2 (Basic quadratic penalty method). Choose $\sigma_0 > 0$ and $\varepsilon_0 > 0$. For $k = 0, 1, 2, \dots$:

1. Find x^k with $\|\nabla \Phi_{\sigma_k}(x^k)\| \leq \varepsilon_k$.
2. Choose $\sigma_{k+1} < \sigma_k$ and $\varepsilon_{k+1} < \varepsilon_k$.

8.2 Convergence of the penalty method

Theorem 8.3 (Global convergence of the quadratic penalty method). *Apply Algorithm 8.2 to (eCP). Assume $f, c \in \mathcal{C}^1$, set $y_i^k = -c_i(x^k)/\sigma_k$, and assume $\|\nabla \Phi_{\sigma_k}(x^k)\| \leq \varepsilon_k$ with $\varepsilon_k \rightarrow 0$ and*

$\sigma_k \rightarrow 0$ as $k \rightarrow \infty$. Suppose further that $x^k \rightarrow x^*$ where $\{\nabla c_i(x^*) : i = 1, \dots, m\}$ are linearly independent. Then x^* is a KKT point of (eCP) and $y^k \rightarrow y^*$, the corresponding Lagrange multiplier vector.

Proof. Differentiating $\Phi_{\sigma_k}(x)$ and substituting $y^k = -c(x^k)/\sigma_k$:

$$\nabla \Phi_{\sigma_k}(x^k) = \nabla f(x^k) + \frac{1}{\sigma_k} J(x^k)^\top c(x^k) = \nabla f(x^k) - J(x^k)^\top y^k. \quad (\diamond)$$

Step 1: $y^k \rightarrow y^*$. LICQ means $J(x^*)$ has full row rank, so the pseudo-inverse $J(x^*)^+ = (J(x^*)J(x^*)^\top)^{-1}J(x^*)$ exists. Since $x^k \rightarrow x^*$ and J is continuous, $J(x^k)^+$ is well-defined and $\|J(x^k)^+\| \leq C$ for large k and some constant C .

Applying $J(x^k)^+$ to both sides of $(\diamond)$ and using $J^+ J^\top = I$:

$$J(x^k)^+ \nabla \Phi_{\sigma_k}(x^k) = J(x^k)^+ \nabla f(x^k) - \underbrace{J(x^k)^+ J(x^k)^\top}_{=I} y^k = J(x^k)^+ \nabla f(x^k) - y^k.$$

Taking norms:

$$\|J(x^k)^+ \nabla f(x^k) - y^k\| = \|J(x^k)^+ \nabla \Phi_{\sigma_k}(x^k)\| \leq \|J(x^k)^+\| \cdot \|\nabla \Phi_{\sigma_k}(x^k)\| \leq C \epsilon_k \rightarrow 0. \quad (\diamond \diamond)$$

Now define $y^* := J(x^*)^+ \nabla f(x^*)$. By the triangle inequality:

$$\|y^k - y^*\| \leq \underbrace{\|y^k - J(x^k)^+ \nabla f(x^k)\|}_{\rightarrow 0 \text{ by } (\diamond \diamond)} + \underbrace{\|J(x^k)^+ \nabla f(x^k) - J(x^*)^+ \nabla f(x^*)\|}_{\rightarrow 0 \text{ by continuity of } J^+ \text{ and } \nabla f}.$$

Hence $y^k \rightarrow y^*$.

Step 2: $c(x^*) = 0$. Note that $c(x^k) = -\sigma_k y^k$. Since $\sigma_k \rightarrow 0$ and $\|y^k\|$ is bounded (as $y^k \rightarrow y^*$), we get $\|c(x^k)\| \rightarrow 0$. By continuity, $c(x^*) = 0$.

Step 3: $\nabla f(x^*) = J(x^*)^\top y^*$. The LHS of $(\diamond)$ satisfies $\|\nabla \Phi_{\sigma_k}(x^k)\| \leq \epsilon_k \rightarrow 0$. The RHS converges to $\nabla f(x^*) - J(x^*)^\top y^*$ by continuity. Equating the limits: $\nabla f(x^*) = J(x^*)^\top y^*$.

Steps 2 and 3 give the KKT conditions at x^* . □

Remark 8.4. If $J(x^*)$ is not full rank then x^* instead (locally) minimizes the infeasibility $\|c(x)\|$.

Remark 8.5 (Ill-conditioning of $\nabla^2 \Phi_{\sigma_k}$). A serious practical drawback of the basic penalty method is that, asymptotically, $\nabla^2 \Phi_{\sigma_k}(x^k)$ has m eigenvalues of order $O(1/\sigma_k)$ and the remaining $n - m$ eigenvalues of order $O(1)$ — so the condition number is $O(1/\sigma_k)$ and *blows up* as $\sigma_k \rightarrow 0$. Consequently we may not be able to compute x^k accurately by Newton's method on Φ_{σ_k} . This motivates augmented Lagrangians (Chapter 9), which avoid the need to drive $\sigma_k \rightarrow 0$.

8.3 Other penalty functions

The quadratic penalty is *smooth* but *inexact*: $\sigma_k \rightarrow 0$ is required, with the ill-conditioning consequences in Remark 8.5. *Exact* penalty functions allow convergence at a fixed (sufficiently small) σ , avoiding ill-conditioning, but are typically non-smooth and so harder to minimise.

Definition 8.6 (Exact penalty function). A penalty function $\Phi(x, \sigma)$ is *exact* if there exists $\sigma^* > 0$ such that for all $\sigma < \sigma^*$, every local solution of the original problem is a local minimizer of $\Phi(\cdot, \sigma)$.

Example 8.7 (ℓ_1 and ℓ_2 penalty functions). For (CP),

$$\Phi_1(x, \sigma) = f(x) + \frac{1}{\sigma} \sum_{i \in E} |c_i(x)| + \frac{1}{\sigma} \sum_{i \in I} [c_i(x)]_-,$$

$$\Phi_2(x, \sigma) = f(x) + \frac{1}{\sigma} \|c_E(x)\|_2 + \frac{1}{\sigma} \|[c_I(x)]_-\|_2.$$

Both are exact (but generally non-smooth); the quadratic penalty is smooth but not exact.

Chapter 9

Augmented Lagrangian Methods

The quadratic penalty method has a fundamental drawback: $\sigma_k \rightarrow 0$ is required, leading to severe ill-conditioning of the subproblem Hessian (Remark 8.5). The augmented Lagrangian method retains the quadratic penalty term but adds an explicit estimate $u \in \mathbb{R}^m$ of the Lagrange multipliers, so that as $u \rightarrow y^*$ feasibility is achieved *without* driving σ to zero.

9.1 The augmented Lagrangian

Definition 9.1 (Augmented Lagrangian). The *augmented Lagrangian* for (eCP) is

$$\Phi(x, u, \sigma) = f(x) - u^\top c(x) + \frac{1}{2\sigma} \|c(x)\|^2, \quad (9.1)$$

where $u \in \mathbb{R}^m$ is an estimate of the Lagrange multipliers and $\sigma > 0$.

Remark 9.2 (Two interpretations of $\Phi(x, u, \sigma)$). $\Phi(x, u, \sigma)$ admits two complementary readings:

- **Lagrangian augmented with a quadratic penalty**, $\mathcal{L}(x, u) + \frac{1}{2\sigma} \|c(x)\|^2$. The quadratic term *convexifies* the Lagrangian locally, fixing the well-known issue that the Lagrangian itself is generally a saddle in (x, y) rather than a minimum in x .
- **Shifted quadratic penalty**: completing the square in $c(x)$ gives $\Phi(x, u, \sigma) = f(x) + \frac{1}{2\sigma} \|c(x) - \sigma u\|^2 - \frac{\sigma}{2} \|u\|^2$, a quadratic penalty centred at the shifted target $c(x) = \sigma u$ rather than $c(x) = 0$. As $u \rightarrow y^*$, the shift vanishes correctly even with σ bounded away from zero.

The aim is to adjust both u and σ to encourage convergence.

Lagrange multiplier estimates. A short calculation gives

$$\nabla_x \Phi(x, u, \sigma) = \nabla f(x) - J(x)^\top y(x), \quad y(x) := u - c(x)/\sigma,$$

so the gradient of the augmented Lagrangian coincides with the gradient of the ordinary Lagrangian evaluated at the multiplier estimate $y(x)$. This is the update rule used in Algorithm 9.3.

9.2 The basic algorithm

Algorithm 9.3 (Augmented Lagrangian method). Choose $u^0 \in \mathbb{R}^m$, $\sigma_0 > 0$, $\varepsilon_0 > 0$. For $k = 0, 1, \dots$:

1. Find x^k with $\|\nabla_x \Phi(x^k, u^k, \sigma_k)\| \leq \varepsilon_k$.
2. Update $u^{k+1} = u^k - c(x^k)/\sigma_k$.
3. Choose $\sigma_{k+1} \leq \sigma_k$, $\varepsilon_{k+1} < \varepsilon_k$.

9.3 Convergence

Theorem 9.4 (Global convergence of augmented Lagrangian). *Apply Algorithm 9.3 to (eCP). Assume $f, c \in \mathcal{C}^1$, let $y^k = u^k - c(x^k)/\sigma_k$, and $\|\nabla_x \Phi(x^k, u^k, \sigma_k)\| \leq \varepsilon_k$ with $\varepsilon_k \rightarrow 0$. Suppose $x^k \rightarrow x^*$ where $\{\nabla c_i(x^*)\}_{i=1}^m$ are linearly independent. Then $y^k \rightarrow y^*$ with $\nabla f(x^*) - J(x^*)^\top y^* = 0$. Moreover, if either $\sigma_k \rightarrow 0$ for bounded u^k , or $u^k \rightarrow y^*$ for bounded σ_k , then x^* is a KKT point of (eCP) with associated multipliers y^* .*

Proof. Differentiating $\Phi_{\sigma_k}(x)$ and substituting $y^k = u^k - c(x^k)/\sigma_k$:

$$\nabla \Phi_{\sigma_k}(x^k) = \nabla f(x^k) - J(x^k)^\top u^k + \frac{1}{\sigma_k} J(x^k)^\top c(x^k) = \nabla f(x^k) - J(x^k)^\top y^k. \quad (\diamond)$$

Step 1: $y^k \rightarrow y^*$. LICQ means $J(x^*)$ has full row rank, so the pseudo-inverse $J(x^*)^+ = (J(x^*)J(x^*)^\top)^{-1}J(x^*)$ exists. Since $x^k \rightarrow x^*$ and J is continuous, $J(x^k)^+$ is well-defined and $\|J(x^k)^+\| \leq C$ for large k and some constant C .

Applying $J(x^k)^+$ to both sides of $(\diamond)$ and using $J^+ J^\top = I$:

$$J(x^k)^+ \nabla \Phi_{\sigma_k}(x^k) = J(x^k)^+ \nabla f(x^k) - \underbrace{J(x^k)^+ J(x^k)^\top}_{=I} y^k = J(x^k)^+ \nabla f(x^k) - y^k.$$

Taking norms:

$$\|J(x^k)^+ \nabla f(x^k) - y^k\| = \|J(x^k)^+ \nabla \Phi_{\sigma_k}(x^k)\| \leq \|J(x^k)^+\| \cdot \|\nabla \Phi_{\sigma_k}(x^k)\| \leq C \varepsilon_k \rightarrow 0. \quad (\diamond\diamond)$$

Now define $y^* := J(x^*)^+ \nabla f(x^*)$. By the triangle inequality:

$$\|y^k - y^*\| \leq \underbrace{\|y^k - J(x^k)^+ \nabla f(x^k)\|}_{\rightarrow 0 \text{ by } (\diamond\diamond)} + \underbrace{\|J(x^k)^+ \nabla f(x^k) - J(x^*)^+ \nabla f(x^*)\|}_{\rightarrow 0 \text{ by continuity of } J^+ \text{ and } \nabla f}.$$

Hence $y^k \rightarrow y^*$.

Step 2: $c(x^*) = 0$. Note that $c(x^k) = \sigma_k(u^k - y^k)$. If u_k is bounded and $\sigma_k \rightarrow 0$, then $c(x^k) \rightarrow 0$. If σ_k is bounded and $u_k \rightarrow y^*$, then $c(x^k) \rightarrow 0$. Hence $c(x^k) \rightarrow 0$, and so $c(x^*) = 0$ by continuity of c .

Step 3: $\nabla f(x^*) = J(x^*)^\top y^*$. The LHS of $(\diamond)$ satisfies $\|\nabla \Phi_{\sigma_k}(x^k)\| \leq \varepsilon_k \rightarrow 0$. The RHS converges to $\nabla f(x^*) - J(x^*)^\top y^*$ by continuity. Equating the limits: $\nabla f(x^*) = J(x^*)^\top y^*$.

Steps 2 and 3 give the KKT conditions at x^* . □

Remark 9.5 (Why the augmented Lagrangian helps). The crucial difference from the basic penalty method is the second alternative in Theorem 9.4: convergence is achieved as soon as $u^k \rightarrow y^*$, with σ_k bounded away from zero. This means the ill-conditioning of the subproblem Hessian is controlled — $\nabla_{xx}^2 \Phi(x, u, \sigma) = \nabla_{xx}^2 \mathcal{L}(x, y) + \frac{1}{\sigma} J(x)^\top J(x)$ has condition number $O(1/\sigma)$, which stays bounded if σ does. Augmented Lagrangian methods underpin the LANCELOT software and are competitive with state-of-the-art nonlinear programming codes.

Chapter 10

Interior Point Methods

For inequality-constrained problems, an alternative to penalty methods is to *stay strictly inside the feasible region throughout* and rely on a *barrier* that diverges as $c_i(x) \downarrow 0$. As we relax the barrier parameter μ , the unconstrained minimisers approach the boundary along the so-called *central path*.

We consider the inequality-constrained problem

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad c(x) \geq 0, \quad (\text{iCP})$$

where $c = (c_1, \dots, c_p) : \mathbb{R}^n \rightarrow \mathbb{R}^p$ is smooth.

10.1 The logarithmic barrier function

Why the logarithm? Since $-\log c_i(x) \rightarrow +\infty$ as $c_i(x) \downarrow 0$, any minimiser $x(\mu)$ of f_μ must lie strictly inside Ω — the constraint $c(x) > 0$ is enforced *implicitly* rather than as an explicit inequality. When μ is large, $x(\mu)$ sits “far inside” Ω ; when μ is small, f dominates and $x(\mu)$ is close to the optimal boundary of Ω (this is also the regime in which ill-conditioning kicks in; see below).

Definition 10.1 (Barrier function). For $\mu > 0$, the *logarithmic barrier function* is

$$f_\mu(x) = f(x) - \mu \sum_{i=1}^p \log c_i(x), \quad (10.1)$$

defined on the strict interior $\{x : c(x) > 0\}$.

Lemma 10.2 (Barrier derivatives). With $J(x)$ the Jacobian of c and $C(x) = \text{diag}(c_1(x), \dots, c_p(x))$,

$$\begin{aligned} \nabla f_\mu(x) &= \nabla f(x) - \mu J(x)^\top C^{-1}(x) = \nabla f(x) - J(x)^\top \lambda(x), \\ \nabla^2 f_\mu(x) &= \nabla^2 f(x) - \sum_{i=1}^p \frac{\mu}{c_i(x)} \nabla^2 c_i(x) + \mu J(x)^\top C(x)^{-2} J(x), \end{aligned}$$

where $\lambda_i(x) := \mu/c_i(x)$ are estimates of the Lagrange multipliers. □

Remark 10.3 (Connection to the Lagrangian). Lemma 10.2 reveals a striking fact: $\nabla f_\mu(x) = \nabla_x \mathcal{L}(x, \lambda(x))$, the gradient of the Lagrangian of (iCP) evaluated at the multiplier estimate $\lambda(x) = \mu/c(x)$. Hence f_μ stationarity coincides with stationarity of the Lagrangian of the original problem, with multipliers $\lambda(x)$. As $\mu \rightarrow 0$ the conditions $\lambda_i(x)c_i(x) = \mu \rightarrow 0$ recover complementarity.

10.2 The central path

Theorem 10.4 (Local existence of the central path). *Assume $\Omega^o \neq \emptyset$ and x^* is a local minimiser of (iCP) satisfying:*

1. $\lambda_i^* > 0$ whenever $c_i(x^*) = 0$ (strict complementarity);
2. $\{\nabla c_i(x^*) : i \in \mathcal{A}\}$ are linearly independent (LICQ);
3. there exists $\alpha > 0$ such that $s^\top \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*) s \geq \alpha \|s\|^2$ for all s with $J_{\mathcal{A}}(x^*)s = 0$.

Then there exists a unique, continuously differentiable path $\{x(\mu) : \mu_\varepsilon > \mu > 0\}$ of (global) minimisers of f_μ , with $x(\mu) \rightarrow x^$ as $\mu \rightarrow 0$.* $\square$

10.3 The basic barrier algorithm

Algorithm 10.5 (Basic barrier algorithm). Choose $\mu_0 > 0$, $\varepsilon_0 > 0$ and x^0 with $c(x^0) > 0$. For $k = 0, 1, \dots$:

1. Find x^k with $c(x^k) > 0$ and $\|\nabla f_{\mu_k}(x^k)\| \leq \varepsilon_k$.
2. Choose $\mu_{k+1} < \mu_k$, $\varepsilon_{k+1} < \varepsilon_k$.

10.4 Convergence

Theorem 10.6 (Global convergence of the barrier method). *Apply Algorithm 10.5 to (iCP). Assume $f, c \in \mathcal{C}^1$, $\lambda_i^k = \mu_k / c_i(x^k)$ for $i = 1, \dots, p$ with $c(x^k) > 0$, $\mu_k > 0$, and $\|\nabla f_{\mu_k}(x^k)\| \leq \varepsilon_k$ with $\varepsilon_k \rightarrow 0$, $\mu_k \rightarrow 0$ as $k \rightarrow \infty$. Suppose $x^k \rightarrow x^*$ where the active gradients $\{\nabla c_i(x^*) : i \in \mathcal{A}\}$ are linearly independent (i.e. LICQ), $\mathcal{A} = \{i : c_i(x^*) = 0\}$. Then x^* is a KKT point of (iCP) and $\lambda^k \rightarrow \lambda^*$, the associated Lagrange multipliers.*

Proof. Let $\mathcal{A} = \{i : c_i(x^*) = 0\}$ and $\mathcal{I} = \{1, \dots, p\} \setminus \mathcal{A}$. The barrier gradient is

$$\nabla f_{\mu_k}(x^k) = \nabla f(x^k) - J_{\mathcal{A}}(x^k)^\top \lambda_{\mathcal{A}}^k - J_{\mathcal{I}}(x^k)^\top \lambda_{\mathcal{I}}^k. \quad (\diamond)$$

Note that $c(x^k) > 0$ and $x^k \rightarrow x^*$ give $c(x^*) \geq 0$ by continuity, so feasibility is satisfied.

Step 1: $\lambda_{\mathcal{I}}^k \rightarrow 0$. For $i \in \mathcal{I}$: $c_i(x^*) > 0$, so $\lambda_i^k = \mu_k / c_i(x^k) \rightarrow 0 / c_i(x^*) = 0$ as $\mu_k \rightarrow 0$.

Step 2: $\lambda_{\mathcal{A}}^k \rightarrow \lambda_{\mathcal{A}}^*$. Rearranging $(\diamond)$ and taking norms:

$$\left\| \nabla f(x^k) - J_{\mathcal{A}}(x^k)^\top \lambda_{\mathcal{A}}^k \right\| \leq \underbrace{\left\| \nabla f_{\mu_k}(x^k) \right\|}_{\leq \varepsilon_k} + \underbrace{\left\| J_{\mathcal{I}}(x^k)^\top \lambda_{\mathcal{I}}^k \right\|}_{\leq \|J_{\mathcal{I}}(x^k)\| \cdot \|\lambda_{\mathcal{I}}^k\|} \rightarrow 0, \quad (\diamond\diamond)$$

using $\varepsilon_k \rightarrow 0$, $\lambda_{\mathcal{I}}^k \rightarrow 0$, and $J_{\mathcal{I}}(x^k)$ bounded.

LICQ gives $J_{\mathcal{A}}(x^k)^+$ well-defined and bounded: $\|J_{\mathcal{A}}(x^k)^+\| \leq C$. Since $J_{\mathcal{A}}^+ J_{\mathcal{A}}^\top = I$, applying $J_{\mathcal{A}}(x^k)^+$ to $\nabla f(x^k) - J_{\mathcal{A}}(x^k)^\top \lambda_{\mathcal{A}}^k$:

$$\left\| J_{\mathcal{A}}(x^k)^+ \nabla f(x^k) - \lambda_{\mathcal{A}}^k \right\| \leq C \cdot \left\| \nabla f(x^k) - J_{\mathcal{A}}(x^k)^\top \lambda_{\mathcal{A}}^k \right\| \rightarrow 0.$$

Define $\lambda_{\mathcal{A}}^* := J_{\mathcal{A}}(x^*)^+ \nabla f(x^*)$. By triangle inequality and continuity ($x^k \rightarrow x^*$):

$$\|\lambda_{\mathcal{A}}^k - \lambda_{\mathcal{A}}^*\| \leq \underbrace{\|\lambda_{\mathcal{A}}^k - J_{\mathcal{A}}(x^k)^+ \nabla f(x^k)\|}_{\rightarrow 0} + \underbrace{\|J_{\mathcal{A}}(x^k)^+ \nabla f(x^k) - J_{\mathcal{A}}(x^*)^+ \nabla f(x^*)\|}_{\rightarrow 0 \text{ by continuity}} \rightarrow 0.$$

Since $\lambda_{\mathcal{A}}^k > 0$ by definition and $\lambda_{\mathcal{A}}^k \rightarrow \lambda_{\mathcal{A}}^*$, we get $\lambda_{\mathcal{A}}^* \geq 0$ (dual feasibility).

Step 3: Stationarity. $(\diamond\diamond)$ gives $\nabla f(x^k) - J_{\mathcal{A}}(x^k)^\top \lambda_{\mathcal{A}}^k \rightarrow 0$. Taking limits: $\nabla f(x^*) = J_{\mathcal{A}}(x^*)^\top \lambda_{\mathcal{A}}^*$.

Conclusion. Set $\lambda_{\mathcal{I}}^* = 0$. Then stationarity $\nabla f(x^*) = J(x^*)^\top \lambda^*$, feasibility $c(x^*) \geq 0$, dual feasibility $\lambda^* \geq 0$, and complementarity $\lambda_i^* c_i(x^*) = 0$ for all i (active: $c_i = 0$; inactive: $\lambda_i^* = 0$) are all satisfied. These are the KKT conditions.

□

Remark 10.7 (Difficulties of basic (primal) IPMs).

- *Ill-conditioning of $\nabla^2 f_\mu$.* As $\mu \rightarrow 0$, $\mu J(x)^\top C(x)^{-2} J(x)$ has $|\mathcal{A}|$ eigenvalues of order $O(1/\mu)$ tending to infinity, while the rest are $O(1)$. The condition number is $O(1/\mu_k)$, blowing up. This is the main reason barrier methods fell out of favour in the 1960s, until primal-dual methods revived them in the 1980s.
- *Poor starting points.* The current iterate x^k is generally a poor starting point for the next subproblem $\min f_{\mu_{k+1}}$: a full Newton step typically violates $c(x^k + s^k) > 0$ as soon as μ is meaningfully decreased, so progress is slow.

The remedy for both is to switch to primal-dual methods.

10.5 Primal-dual interior point methods

Remark 10.8 (Primal-dual reformulation). Modern interior point software (KNITRO, IPOPT, GALAHAD) works with the *perturbed KKT system*

$$\nabla f(x) - J(x)^\top \lambda = 0, \quad C(x)\lambda = \mu e, \quad c(x) > 0, \quad \lambda > 0,$$

where $e = (1, \dots, 1)^\top$. The path traced by solutions as $\mu \downarrow 0$ is the *central path*. Newton's method applied to the perturbed KKT system treats (x, λ) as joint variables — multiplier updates are computed explicitly each iteration, in contrast to the implicit estimate $\lambda(x) = \mu/c(x)$ in the primal method.

Benefits over the primal barrier method:

- Ill-conditioning is sidestepped: solving the linear system in the right (x, λ) subspaces, the asymptotic conditioning is benign.
- Current iterates are good starting points for the next subproblem, so only a few Newton iterations per μ_k are needed.
- Aggressive parameter schedule $\mu_{k+1} = O(\mu_k^2)$ yields *superlinear* asymptotic convergence.

Computing an initial x^0 with $c(x^0) > 0$ is itself nontrivial; various heuristics exist (e.g. phase-I problems).

Remark 10.9 (Linear programming: simplex vs. IPMs). For LP, the simplex method has worst-case exponential complexity (Klee–Minty 1972) but very good average-case behaviour. Karmarkar’s IPM (1984) gave the first practical polynomial-time algorithm for LP; the central path is unique and global, and Newton’s method on the barrier function can be precisely quantified. IPMs scale to very large LPs, with numerically observed average complexity $O(\log(\text{LP dim}))$ iterations, each more expensive than a simplex pivot.

Appendix A

Useful background

A.1 Taylor's theorem

Theorem A.1 (First-order Taylor). *Let $f \in \mathcal{C}^1$. For any $x, s \in \mathbb{R}^n$ there exists $\tilde{\alpha} \in (0, 1)$ with $f(x + s) = f(x) + \nabla f(x + \tilde{\alpha}s)^\top s$.*

Theorem A.2 (Second-order Taylor). *Let $f \in \mathcal{C}^2$. For any $x, s \in \mathbb{R}^n$ there exists $\tilde{\alpha} \in (0, 1)$ with*

$$f(x + s) = f(x) + \nabla f(x)^\top s + \frac{1}{2}s^\top \nabla^2 f(x + \tilde{\alpha}s)s.$$

A.2 Lipschitz continuity

Definition A.3. A function $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is *Lipschitz continuous* on $\mathbb{R}^n$ with constant $L > 0$ if $\|g(y) - g(x)\| \leq L \|y - x\|$ for all $x, y \in \mathbb{R}^n$.

A.3 Convexity

Theorem A.4 (First-order characterization). *Let $f \in \mathcal{C}^1$. Then f is convex if and only if $f(y) \geq f(x) + \nabla f(x)^\top (y - x)$ for all x, y .*

Theorem A.5 (Second-order characterization). *Let $f \in \mathcal{C}^2$. Then f is convex if and only if $\nabla^2 f(x) \succeq 0$ for all x .*

Proof. ($\Leftarrow$) Suppose $\nabla^2 f(x) \succeq 0$ for all x . Fix $x, y \in \mathbb{R}^n$ and put $s := y - x$. By second-order Taylor's theorem with remainder, there exists $\tilde{\alpha} \in (0, 1)$ such that

$$f(y) = f(x + s) = f(x) + \nabla f(x)^\top s + \frac{1}{2}s^\top \nabla^2 f(x + \tilde{\alpha}s)s.$$

Since $\nabla^2 f(x + \tilde{\alpha}s) \succeq 0$, the last term is nonnegative, giving $f(y) \geq f(x) + \nabla f(x)^\top (y - x)$. This is the first-order characterisation, so f is convex.

($\Rightarrow$) Suppose f is convex. Fix $x, d \in \mathbb{R}^n$ and $t > 0$. The first-order characterisation gives both $f(x + td) \geq f(x) + t \nabla f(x)^\top d$ and $f(x) \geq f(x + td) + (-td)^\top \nabla f(x + td)$. Adding:

$$0 \geq t[\nabla f(x) - \nabla f(x + td)]^\top d.$$

Dividing by $t^2 > 0$ and letting $t \downarrow 0$: $0 \leq \frac{[\nabla f(x+td) - \nabla f(x)]^\top d}{t} \rightarrow d^\top \nabla^2 f(x)d$. Hence $d^\top \nabla^2 f(x)d \geq 0$ for all d , so $\nabla^2 f(x) \succeq 0$. $\square$